

# **Microscopic Foundations of Relativistic Fluid Dynamics**

Dirk H. Rischke

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# Contents

|          |                                                                   |           |
|----------|-------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Relativistic Fluid Dynamics</b>                                | <b>1</b>  |
| 1.1      | Thermodynamics                                                    | 2         |
| 1.2      | Relativistic ideal fluid dynamics                                 | 4         |
| 1.2.1    | Conserved currents in an ideal fluid                              | 4         |
| 1.2.2    | Equations of motion                                               | 6         |
| 1.2.3    | Covariant thermodynamics and entropy production                   | 7         |
| 1.3      | Relativistic dissipative fluid dynamics                           | 8         |
| 1.3.1    | Matching conditions                                               | 10        |
| 1.3.2    | Tensor decomposition of $\tau^{\mu\nu}$                           | 11        |
| 1.3.3    | Definition of the local rest frame and equations of motion        | 12        |
| 1.3.4    | Relativistic Navier-Stokes theory                                 | 14        |
| 1.3.5    | Gradient expansion and Navier-Stokes theory                       | 15        |
| 1.4      | Causal fluid dynamics                                             | 18        |
| 1.4.1    | Diffusion equation and acausality in heat conduction              | 18        |
| 1.4.2    | Transient theory of fluid dynamics                                | 20        |
| 1.5      | Transient thermodynamics and Israel-Stewart theory                | 22        |
| <b>2</b> | <b>Linear Stability and Causality</b>                             | <b>27</b> |
| 2.1      | Fluid-dynamical equations linearized around global equilibrium    | 28        |
| 2.2      | Linearized fluid-dynamical equations in Fourier space             | 31        |
| 2.2.1    | Tensor decomposition in Fourier space                             | 31        |
| 2.2.2    | Longitudinal and transverse components                            | 32        |
| 2.3      | Ideal fluid dynamics                                              | 34        |
| 2.4      | Relativistic Navier-Stokes theory                                 | 36        |
| 2.4.1    | Transverse modes                                                  | 37        |
| 2.4.2    | Longitudinal modes                                                | 38        |
| 2.4.3    | Causality and stability of Navier-Stokes theory                   | 40        |
| 2.5      | Transient theory of fluid dynamics                                | 41        |
| 2.5.1    | Transverse modes in the rest frame                                | 42        |
| 2.5.2    | Longitudinal modes in the rest frame                              | 45        |
| 2.5.3    | Stability for a moving background                                 | 48        |
| 2.5.4    | Causality of wave propagation                                     | 51        |
| 2.6      | Summary                                                           | 52        |
| <b>3</b> | <b>Fluid Dynamics from Kinetic Theory: Traditional Approaches</b> | <b>57</b> |
| 3.1      | Matching fluid-dynamical with kinetic degrees of freedom          | 57        |
| 3.1.1    | Macroscopic conservation laws                                     | 58        |
| 3.1.2    | Fluid-dynamical variables and matching conditions                 | 59        |

|          |                                                                                   |            |
|----------|-----------------------------------------------------------------------------------|------------|
| 3.2      | Chapman-Enskog theory . . . . .                                                   | 61         |
| 3.2.1    | Solving the Chapman-Enskog expansion: zeroth- and first-order solutions . . . . . | 65         |
| 3.2.2    | Minimal truncation scheme . . . . .                                               | 70         |
| 3.3      | Israel-Stewart theory . . . . .                                                   | 73         |
| 3.3.1    | 14-moment approximation . . . . .                                                 | 73         |
| 3.3.2    | Matching procedure . . . . .                                                      | 75         |
| 3.3.3    | Moment equations . . . . .                                                        | 77         |
| 3.3.4    | Calculation of the collision integrals . . . . .                                  | 79         |
| 3.3.5    | Hydrodynamic equations of motion . . . . .                                        | 81         |
| 3.4      | Summary . . . . .                                                                 | 83         |
| <b>4</b> | <b>Method of Moments</b>                                                          | <b>87</b>  |
| 4.1      | Moment expansion . . . . .                                                        | 88         |
| 4.2      | Equations of motion for the irreducible moments . . . . .                         | 92         |
| 4.3      | Collision term . . . . .                                                          | 94         |
| 4.4      | Summary . . . . .                                                                 | 97         |
| 4.5      | Appendix 1: Irreducible projection operators . . . . .                            | 97         |
| 4.6      | Appendix 2: Thermodynamic integrals and properties . . . . .                      | 99         |
| 4.7      | Appendix 3: Orthogonality of the irreducible tensors . . . . .                    | 102        |
| 4.8      | Appendix 4: Orthogonal polynomials . . . . .                                      | 103        |
| <b>5</b> | <b>Fluid Dynamics from the Method of Moments</b>                                  | <b>107</b> |
| 5.1      | Power counting . . . . .                                                          | 108        |
| 5.2      | Resummed transient relativistic fluid dynamics . . . . .                          | 110        |
| 5.3      | Resummed transient relativistic fluid dynamics: 14 dynamical variables . . . . .  | 116        |
| 5.4      | Transport coefficients . . . . .                                                  | 118        |
| 5.4.1    | 14-moment approximation . . . . .                                                 | 119        |
| 5.4.2    | 23-moment approximation and beyond . . . . .                                      | 120        |
| 5.5      | Discussion: Navier-Stokes limit and causality . . . . .                           | 122        |
| 5.6      | Resummed transient relativistic fluid dynamics: 23 dynamical variables . . . . .  | 123        |
| 5.7      | Comparisons with microscopic theory . . . . .                                     | 126        |
| 5.7.1    | Stationary shock solutions . . . . .                                              | 127        |
| 5.8      | Summary . . . . .                                                                 | 130        |
| 5.9      | Appendix 1: Transport coefficients in Eq. (5.44) . . . . .                        | 134        |
| 5.10     | Appendix 2: Calculation of the collision integrals . . . . .                      | 136        |
| 5.10.1   | Particle-diffusion current . . . . .                                              | 138        |
| 5.10.2   | Shear-stress tensor . . . . .                                                     | 139        |
| 5.11     | Appendix 3: Calculation of $\gamma_1^{(2)}$ . . . . .                             | 140        |
| 5.11.1   | 14-moment approximation . . . . .                                                 | 141        |
| 5.11.2   | 23-moment approximation . . . . .                                                 | 141        |
| 5.12     | Appendix 4: Transport coefficients in Sec. 5.6 . . . . .                          | 142        |

# 1 Relativistic Fluid Dynamics

The physical description of a system consisting of many degrees of freedom is in general quite complicated. However, if one is interested in the **large-distance, long-timescale** behavior of the system, it becomes possible to devise an **effective theory**, taking into account only the degrees of freedom that are relevant on these scales. This happens because, on macroscopic time and length scales, we are not able to observe the microscopic degrees of freedom of the underlying theory, but only average quantities resulting from interactions on the microscopic level. Most of the microscopic quantities vary rapidly in space and time, leading to very small changes of the average values, and are not expected to contribute to the macroscopic dynamics. On the other hand, the few variables that do vary **slowly**, such as **conserved quantities**, are expected to be relevant for the **effective description** of the system on **macroscopic length and time scales**.

**Fluid dynamics** is a typical example of such an effective theory. It is a **classical field theory** that describes the **macroscopic dynamics** of systems called **fluids**. A fluid is a **continuous system** in which every infinitesimal volume element is (usually) assumed to be **close to thermodynamic equilibrium** and to maintain the proximity to equilibrium throughout its evolution. In other words, in the vicinity of each point in space, we define an infinitesimal volume, called fluid element, in which the matter is taken to be homogeneous, i.e., any spatial gradients can be neglected, and is described by a finite set of thermodynamic variables and currents. This means that each fluid element must be **large enough relative to the microscopic length scales**, to guarantee the proximity to thermodynamic equilibrium, and, at the same time, it must be **small enough relative to the macroscopic length scales**, to ensure the continuum limit.

At first glance, the simultaneous applicability of the continuous (zero fluid-element volume) and thermodynamic (infinite fluid-element volume) limits might seem contradictory. However, if the microscopic and macroscopic scales of the system are **sufficiently far separated**, it is always possible to ensure the existence of a volume that is small when compared with the macroscopic scales, and large when compared with the microscopic ones. For example, considering the case of water, a fluid element with a volume of about  $1 \text{ mm}^3$  is small enough to assure the continuous-limit approximation and large enough to enclose many molecules and to apply the thermodynamic limit. Note that, for small or rapidly changing systems, such a separation of scales may not be so clear, making it difficult to ensure the proximity to local thermal equilibrium. This is of particular importance when we apply fluid dynamics to describe the hot and dense matter formed in relativistic heavy-ion collisions, where a clear separation between microscopic and macroscopic scales does not exist. The progress in developing dissipative theories of relativistic fluid dynamics achieved in recent years has been largely driven by applying them to describe the dynamical evolution of heavy-ion collisions.

In this chapter, we discuss the basic aspects of relativistic fluid dynamics from a phenomenological perspective. It is organized as follows: In Sec. 1.1, we introduce the basic laws of thermodynamics and derive the thermodynamic relations that are useful for this book. Section 1.2 contains a brief review of relativistic ideal fluid dynamics. We derive the general form of the conserved currents of an ideal fluid and their equations of motion. Then, Sec. 1.3 shows how to introduce dissipation in fluid dynamics. Here, we explain the basic aspects of dissipative fluid dynamics and derive a covariant version of Navier-Stokes theory using the second law of thermodynamics and, also, via the gradient expansion. In Sec. 1.4 we review the problems of Navier-Stokes theory in the relativistic regime, i.e., the acausality and instability of this theory. We also explain how to render Navier-Stokes theory causal and stable, and to derive a consistent theory of fluid dynamics. Finally, in Sec. 1.5, we discuss Israel-Stewart theory and show how to derive causal fluid-dynamical equations from the second law of thermodynamics.

### 1.1 Thermodynamics

Thermodynamics is a theory empirically constructed to describe the thermodynamical-equilibrium state in macroscopic systems. It attempts to describe such a state in terms of a small set of **extensive quantities**, such as the total **energy**,  $E$ , **volume**,  $V$ , and (net) **number of particles**,  $N$ , of the system. Thermodynamics is based on four phenomenological laws, obtained over the years by experimental observation [1]:

**Zeroth Law:** Two systems that are in equilibrium with a third system are in equilibrium with each other.

**First Law:** Energy is conserved.

**Second Law:** The change in entropy of a closed thermodynamic system is always positive semi-definite.

**Third Law:** The difference in entropy between systems connected by a reversible process is zero in the limit of vanishing temperature.

In this lecture series, we shall make use of the **first** and **second laws of thermodynamics** and, therefore, it is convenient to discuss them in more detail. The **first law of thermodynamics** implies that small variations of the state variables,  $E$ ,  $V$ , and  $N$ , must be related,

$$\delta E = \delta Q - P\delta V + \mu\delta N , \quad (1.1)$$

where  $P$  and  $\mu$  are the **pressure** and **chemical potential**, respectively. As a conservation law, the first law of thermodynamics postulates that changes in the total energy of the system ( $\delta E$ ) must result from mechanical work done by an external force ( $-P\delta V$ ), from particle exchange with an external medium ( $\mu\delta N$ ), or/and from heat exchange ( $\delta Q$ ). The heat exchange takes into account the energy variations due to changes of internal degrees of freedom that are not described by the state variables. The heat itself is not a state variable since it can depend on the past evolution of the system and may take several values for the same thermodynamic state.

However, when dealing with time-reversible processes, it becomes possible to assign a state variable related to heat. This variable is the **entropy**,  $S$ , and is defined in terms of the heat exchange as  $\delta Q = T\delta S$ , with the **temperature**  $T$  being the proportionality constant. Then, when considering variations between equilibrium states that are infinitesimally close to each other, it is possible to write the first law of thermodynamics in terms of complete differentials of the state variables,

$$dE = TdS - PdV + \mu dN . \quad (1.2)$$

Using Eq. (1.2), it is possible to identify the **intensive quantities**,  $T$ ,  $\mu$ , and  $P$  as the following partial derivatives of the energy,

$$T = \left. \frac{\partial E}{\partial S} \right|_{N,V} , \quad -P = \left. \frac{\partial E}{\partial V} \right|_{S,N} , \quad \mu = \left. \frac{\partial E}{\partial N} \right|_{S,V} . \quad (1.3)$$

The first law of thermodynamics can also be written in terms of entropy variations, i.e.,

$$dS = \frac{1}{T}dE + \frac{P}{T}dV - \frac{\mu}{T}dN , \quad (1.4)$$

in which case the intensive variables can be obtained from partial derivatives of the entropy,

$$\frac{1}{T} = \left. \frac{\partial S}{\partial E} \right|_{N,V} , \quad \frac{P}{T} = \left. \frac{\partial S}{\partial V} \right|_{E,N} , \quad \frac{\mu}{T} = - \left. \frac{\partial S}{\partial N} \right|_{E,V} . \quad (1.5)$$

In the **thermodynamical limit**, the entropy is an **extensive** and **additive** function of the state variables,

$$\lambda S = S(\lambda E, \lambda V, \lambda N) . \quad (1.6)$$

Using this property, it is straightforward to prove that

$$S = \frac{\partial}{\partial \lambda} (\lambda S) = \left. \frac{\partial S}{\partial(\lambda E)} \right|_{\lambda N, \lambda V} E + \left. \frac{\partial S}{\partial(\lambda V)} \right|_{\lambda E, \lambda N} V + \left. \frac{\partial S}{\partial(\lambda N)} \right|_{\lambda E, \lambda V} N , \quad (1.7)$$

which holds for any value of  $\lambda$ . Taking  $\lambda = 1$  and using Eq. (1.5), we derive the so-called **Euler relation**,

$$TS = E + PV - \mu N , \quad (1.8)$$

and using Euler's relation, combined with the first law of thermodynamics, we obtain the **Gibbs-Duhem relation**,

$$VdP = SdT + Nd\mu . \quad (1.9)$$

Together, Eqs. (1.2), (1.8), and (1.9) allow us to derive the thermodynamic relations satisfied by the energy, entropy, and (net) particle number **densities**,  $\varepsilon \equiv E/V$ ,  $s \equiv S/V$ , and  $n \equiv N/V$ , respectively. They are

$$\varepsilon + P = Ts + \mu n , \quad (1.10)$$

$$ds = \beta d\varepsilon - \alpha dn , \quad (1.11)$$

$$dP = sdT + nd\mu , \quad (1.12)$$

where we defined the **inverse temperature**  $\beta = 1/T$  and the **thermal potential**  $\alpha = \mu/T$ . Then, Eqs. (1.11) and (1.12) can be used to derive the relations between the intensive variables and the densities,

$$\beta = \left. \frac{\partial s}{\partial \varepsilon} \right|_n, \quad \alpha = \left. \frac{\partial s}{\partial n} \right|_\varepsilon, \quad s = \left. \frac{\partial P}{\partial T} \right|_\mu, \quad n = \left. \frac{\partial P}{\partial \mu} \right|_T. \quad (1.13)$$

The **second law of thermodynamics** dictates that the entropy of an isolated system must either increase or remain constant. This implies that, if a given system is in **thermodynamic equilibrium**, i.e., if it is in a quasi-stationary state where its extensive and intensive variables no longer change, the **entropy** of this system must remain **constant** (as long as the boundary conditions imposed on the system remain fixed). On the other hand, the entropy of a system that is **out of equilibrium** must always **increase**. This is a very useful and powerful concept that will be extensively used in this chapter. As we will show later, the second law of thermodynamics can even be used to constrain and, sometimes, derive the equations of motion of a viscous fluid.

These are the basic aspects of thermodynamics that we wanted to address (for a more detailed review, see Ref. [1]). It is worth to point out that, although the thermodynamic relations specify how the macroscopic variables are related and how they change with time, they are not enough to extract the **explicit form of the equation of state**, i.e., of the function  $s(\varepsilon, n)$ . In order to determine the entropy density as a function of the state variables  $\varepsilon, n$ , a **microscopic description** of the matter is required, which can only be obtained from a more fundamental approach, such as statistical mechanics.

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**Exercise 1.1:** Prove Eqs. (1.10) – (1.12).

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## 1.2 Relativistic ideal fluid dynamics

We start our discussion of relativistic fluid dynamics by considering the most simple example: the case of an ideal fluid [2, 3]. An ideal fluid is defined by the assumption of **local thermodynamical equilibrium**, i.e., that all fluid elements must be in thermodynamic equilibrium, but not necessarily in the same thermodynamic-equilibrium state (if they are, one speaks of **global** thermodynamical equilibrium). This means that all thermodynamic state variables are functions of the space-time 4-vector  $x^\mu \equiv X = (t, \mathbf{x})^T$ , e.g. temperature  $T(X)$ , chemical potential  $\mu(X)$ . In addition, the fluid is described by a collective velocity field, which is also function of space-time,  $\mathbf{u}(X)$ . From now on, the fields  $T$ ,  $\mu$ , and  $\mathbf{u}$  shall be referred to as **primary fluid-dynamical variables**.

### 1.2.1 Conserved currents in an ideal fluid

The state of a fluid is specified by the densities and currents associated to **conserved quantities**, i.e., **energy**, **momentum**, and (net) **particle number**. For a relativistic fluid, the state variables are the **energy-momentum tensor**,  $T^{\mu\nu}(X)$ , and the (net)

**particle 4-current**,  $N^\mu(X)$ . For an **ideal fluid**, the general form of these currents can be obtained by performing a **Lorentz transformation** to the **local rest frame** of the fluid, in which  $\mathbf{u}(X) = 0$ . In this frame, the energy-momentum tensor,  $T_{\text{RF}}^{\mu\nu}$  (the subscript RF indicates the local rest-frame form of this tensor), should have the characteristic form of a system in **static equilibrium**,

$$T_{\text{RF}}^{\mu\nu} = \begin{pmatrix} \varepsilon & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}, \quad (1.14)$$

i.e., in this frame there is **no flow of energy** ( $T_{\text{RF}}^{i0} = 0$ ), and the force per surface element is **isotropic** and equal to the **thermodynamic pressure** ( $T_{\text{RF}}^{ij} = \delta^{ij}P$ ). In the local rest frame, there should also be **no flow of particles and entropy** and, consequently, the (net) particle and entropy 4-currents in this frame,  $N_{\text{RF}}^\mu$  and  $S_{\text{RF}}^\mu$ , respectively, take the following simple form

$$N_{\text{RF}}^\mu = (n, 0, 0, 0)^T, \quad (1.15)$$

$$S_{\text{RF}}^\mu = (s, 0, 0, 0)^T. \quad (1.16)$$

The form of these tensors in a **general frame**, i.e., a frame where the fluid moves with velocity  $+\mathbf{u}$ , can be derived by applying a Lorentz boost with velocity  $-\mathbf{u}$  to  $N_{\text{RF}}^\mu$ ,  $S_{\text{RF}}^\mu$ , and  $T_{\text{RF}}^{\mu\nu}$ ,

$$N^\mu = \Lambda_\alpha^\mu(-\mathbf{u}) N_{\text{RF}}^\alpha, \quad (1.17)$$

$$S^\mu = \Lambda_\alpha^\mu(-\mathbf{u}) S_{\text{RF}}^\alpha, \quad (1.18)$$

$$T^{\mu\nu} = \Lambda_\alpha^\mu(-\mathbf{u}) \Lambda_\beta^\nu(-\mathbf{u}) T_{\text{RF}}^{\alpha\beta}, \quad (1.19)$$

where we remember that the general form of a Lorentz boost is

$$\Lambda_\alpha^\mu(\mathbf{u}) = \begin{pmatrix} \gamma & -u^x & -u^y & -u^z \\ -u^x & 1 + (\gamma + 1)^{-1} u^x u^x & (\gamma + 1)^{-1} u^x u^y & (\gamma + 1)^{-1} u^x u^z \\ -u^y & (\gamma + 1)^{-1} u^y u^x & 1 + (\gamma + 1)^{-1} u^y u^y & (\gamma + 1)^{-1} u^y u^z \\ -u^z & (\gamma + 1)^{-1} u^z u^x & (\gamma + 1)^{-1} u^z u^y & 1 + (\gamma + 1)^{-1} u^z u^z \end{pmatrix}, \quad (1.20)$$

with  $\gamma = \sqrt{1 + \mathbf{u} \cdot \mathbf{u}}$  being the Lorentz gamma factor. The boost velocity is  $-\mathbf{u}$ , because an observer who sees the fluid move with velocity  $+\mathbf{u}$  moves itself with velocity  $-\mathbf{u}$  with respect to the fluid rest frame.

Then, using a covariant notation, the **conserved currents of an ideal fluid** can be expressed as

$$N_{\text{ideal}}^\mu \equiv N_{(0)}^\mu = n u^\mu, \quad (1.21)$$

$$S_{\text{ideal}}^\mu \equiv S_{(0)}^\mu = s u^\mu, \quad (1.22)$$

$$T_{\text{ideal}}^{\mu\nu} \equiv T_{(0)}^{\mu\nu} = \varepsilon u^\mu u^\nu - \Delta^{\mu\nu} P, \quad (1.23)$$

where  $u^\mu$  is the **velocity 4-vector**,

$$u^\mu = (\gamma, \mathbf{u})^T, \quad (1.24)$$



Note that the velocity 4-vector is constructed to satisfy the normalization condition  $u^\mu u_\mu = 1$ , and, therefore, has only **three independent components**. Also, in Eq. (1.23), we introduced the **projection operator** onto the **3-space orthogonal** to  $u^\mu$ ,

$$\Delta^{\mu\nu} = g^{\mu\nu} - u^\mu u^\nu, \quad (1.25)$$

where  $g_{\mu\nu}$  is the space-time metric (in flat space).  $\Delta^{\mu\nu}$  satisfies all the properties expected of a projector,

$$u_\mu \Delta^{\mu\nu} = u_\nu \Delta^{\mu\nu} = 0, \quad \Delta^\mu_\lambda \Delta^{\lambda\nu} = \Delta^{\mu\nu}, \quad (1.26)$$

and has the following trace,

$$\Delta^\mu_\mu = 3. \quad (1.27)$$

---

**Exercise 1.2:** Prove Eqs. (1.21) – (1.23) by explicit computation of the right-hand sides of Eqs. (1.17) – (1.19).

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## 1.2.2 Equations of motion

The dynamical description of an ideal fluid is obtained using the **conservation laws** of **energy**, **momentum**, and (net) **particle number**. These conservation laws can be mathematically expressed in terms of the following five **continuity equations**,

$$\partial_\mu N_{(0)}^\mu = 0, \quad (1.28)$$

$$\partial_\mu T_{(0)}^{\mu\nu} = 0. \quad (1.29)$$

The partial derivative  $\partial_\mu \equiv \partial/\partial x^\mu$  transforms as a covariant vector under Lorentz transformations and, therefore, Eq. (1.29) transforms as a contravariant 4-vector. As a 4-vector, it is convenient to decompose this equation into a part **parallel** and a part **orthogonal** to  $u^\mu$ . The component parallel to the velocity is obtained by contracting the equation of motion with  $u^\mu$ ,  $u_\alpha \partial_\beta T_{(0)}^{\alpha\beta}$ , while the component orthogonal to the velocity is obtained by contracting it with  $\Delta^{\mu\nu}$ ,  $\Delta^\mu_\alpha \partial_\beta T_{(0)}^{\alpha\beta}$ . This, together with the conservation law for (net) particle number, leads to the **equations of motion of ideal fluid dynamics**,

$$u_\alpha \partial_\beta T_{(0)}^{\alpha\beta} = \dot{\varepsilon} + (\varepsilon + P)\theta = 0, \quad (1.30)$$

$$\Delta^\mu_\alpha \partial_\beta T_{(0)}^{\alpha\beta} = (\varepsilon + P)\dot{u}^\mu - \nabla^\mu P = 0, \quad (1.31)$$

$$\partial_\mu N_{(0)}^\mu = \dot{n} + n\theta = 0, \quad (1.32)$$

where we introduced the **comoving derivative**  $u^\mu \partial_\mu A \equiv \dot{A}$  of any quantity  $A$  and the **space-like gradient**  $\Delta^\lambda_\mu \partial_\lambda \equiv \nabla_\mu$ . We further defined the **expansion scalar**,  $\theta$  as the 4-divergence of the 4-velocity,

$$\theta \equiv \nabla_\mu u^\mu. \quad (1.33)$$

---

**Exercise 1.3:** Prove Eqs. (1.30) – (1.32) by inserting Eqs. (1.21), (1.23) into the conservation laws (1.28), (1.29) and, in the case of energy-momentum conservation, performing the projections onto  $u_\alpha$  and  $\Delta_\alpha^\mu$ .

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Note that an ideal fluid is described by four fields,  $\varepsilon$ ,  $P$ ,  $n$ , and  $u^\mu$ , that contain, in total, **six independent degrees of freedom**. The conservation laws, on the other hand, provide only **five equations of motion**. To **close** this system of equations, we must specify the **equation of state of the fluid**, which gives the pressure as a function of the other thermodynamic variables,

$$P = P(\varepsilon, n) . \quad (1.34)$$

The assumption of local thermal equilibrium guarantees the existence of this function and, hence, assures that the equations of ideal fluid dynamics are always closed. In ideal fluid dynamics, the equation of state essentially defines the type of fluid that is being described – it is the only place where (some of) the microscopic properties of the system must be taken into account.

### 1.2.3 Covariant thermodynamics and entropy production

Using the conserved currents,  $N_{(0)}^\mu$ ,  $S_{(0)}^\mu$ , and  $T_{(0)}^{\mu\nu}$ , we can re-write the equilibrium thermodynamic relations derived in Sec. 1.1, Eqs. (1.10), (1.11), and (1.12), in a covariant form [5, 6, 7]. For this purpose, it is convenient to introduce the following 4-vector,

$$\beta^\mu = \frac{u^\mu}{T} . \quad (1.35)$$

Then, following Israel and Stewart [5, 6, 7], we postulate a **covariant version** of the Gibbs-Duhem relation,

$$d(P\beta^\mu) = N_{(0)}^\mu d\alpha - T_{(0)}^{\mu\nu} d\beta_\nu , \quad (1.36)$$

and of Euler's relation,

$$S_{(0)}^\mu = P\beta^\mu + T_{(0)}^{\mu\nu}\beta_\nu - \alpha N_{(0)}^\mu . \quad (1.37)$$

Equations (1.36) and (1.37) can then be used to derive a covariant form of the first law of thermodynamics (1.11),

$$dS_{(0)}^\mu = \beta_\nu dT_{(0)}^{\mu\nu} - \alpha dN_{(0)}^\mu . \quad (1.38)$$

The above covariant thermodynamic relations were constructed in a such a way that, when contracted with the fluid 4-velocity, the usual thermodynamic relations, Eqs. (1.36), (1.37), and (1.38), are recovered,

$$u_\mu \left( d(P\beta^\mu) - N_{(0)}^\mu d\alpha + T_{(0)}^{\mu\nu} d\beta_\nu \right) = d(P\beta) - nd\alpha + \varepsilon d\beta = 0 , \quad (1.39)$$

$$u_\mu \left( S_{(0)}^\mu - P\beta^\mu - T_{(0)}^{\mu\nu}\beta_\nu + \alpha N_{(0)}^\mu \right) = s + \alpha n - \beta(\varepsilon + P) = 0 , \quad (1.40)$$

$$u_\mu \left( dS_{(0)}^\mu - \beta_\nu dT_{(0)}^{\mu\nu} + \alpha dN_{(0)}^\mu \right) = ds - \beta d\varepsilon + \alpha dn = 0 , \quad (1.41)$$

where we used that  $u_\mu du^\mu = 0$ . Note also that the covariant thermodynamic relations do not contain more information than the usual thermodynamic relations. The projection of Eqs. (1.36), (1.37) and (1.38) onto the 3-space orthogonal to  $u^\mu$  just leads to trivial relations,

$$\Delta_\mu^\alpha \left( d(P\beta^\mu) - N_{(0)}^\mu d\alpha_0 + T_{(0)}^{\mu\nu} d\beta_\nu \right) = 0 \quad \Longrightarrow \quad 0 = 0, \quad (1.42)$$

$$\Delta_\mu^\alpha \left( S_{(0)}^\mu - P\beta^\mu - T_{(0)}^{\mu\nu} \beta_\nu + \alpha N_{(0)}^\mu \right) = 0 \quad \Longrightarrow \quad 0 = 0, \quad (1.43)$$

$$\Delta_\mu^\alpha \left( dS_{(0)}^\mu - \beta_\nu dT_{(0)}^{\mu\nu} + \alpha dN_{(0)}^\mu \right) = 0 \quad \Longrightarrow \quad 0 = 0. \quad (1.44)$$

The first law of thermodynamics, Eq. (1.38), leads to the following equation of motion for the entropy 4-current,

$$\partial_\mu S_{(0)}^\mu = \beta_\nu \partial_\mu T_{(0)}^{\mu\nu} - \alpha \partial_\mu N_{(0)}^\mu, \quad (1.45)$$

which, using the conservation of (net) particle number, energy, and momentum,  $\partial_\mu N_{(0)}^\mu = \partial_\mu T_{(0)}^{\mu\nu} = 0$ , implies the **conservation of entropy**,

$$\partial_\mu S_{(0)}^\mu = 0. \quad (1.46)$$

Note that the entropy conservation appeared naturally, as a consequence of (net) particle number and energy-momentum conservation and the first law of thermodynamics. The equation of motion for the entropy density comes directly from Eq. (1.46),

$$\partial_\mu S_{(0)}^\mu = u^\mu \partial_\mu s + s \partial_\mu u^\mu = \dot{s} + s\theta = 0. \quad (1.47)$$

In this section, we introduced and derived the equations of motion of ideal fluid dynamics. This was done using the conservation laws expected to be valid in a fluid and the assumption of local thermal equilibrium. In the next sections, we show how to introduce dissipation into this scheme.

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**Exercise 1.4:** Convince yourself of the validity of Eqs. (1.39) – (1.44) by explicit calculation.

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## 1.3 Relativistic dissipative fluid dynamics

Relativistic ideal fluid dynamics was derived using the conservation laws, the properties of the Lorentz transformation, and, most importantly, by imposing local thermodynamic equilibrium. While the conservation laws and the properties of the Lorentz transformation are always valid, the assumption of local thermodynamic equilibrium is very restrictive and is never realized in practice. Strictly speaking, a fluid can never maintain exact local thermodynamic equilibrium during the whole course of its dynamical evolution. In this section, we consider a more general theory of fluid dynamics, which attempts to take into account the dissipative processes that must happen in a fluid.

Dissipative effects originate from irreversible thermodynamic processes that occur during the motion of the fluid. In general, each fluid element is not in equilibrium with the rest of the fluid and, in order to approach global equilibrium, it will exchange heat with its surroundings. Furthermore, the fluid elements are in relative motion and can also dissipate energy by friction. All these processes must be included in order to obtain a reasonable description of a relativistic fluid.

The first to propose a covariant formulation of dissipative fluid dynamics were Eckart [4], in 1940, and, later, Landau and Lifshitz [2], in 1959. Both theories, often called **first-order theories**, are based on a **covariant formulation of Navier-Stokes theory**. At that time, Navier-Stokes theory had already become a successful theory of fluid dynamics, being able to describe a wide variety of non-relativistic fluids, from weakly coupled gases, such as air, to strongly coupled fluids, such as water. Therefore, a relativistic extension of Navier-Stokes theory was considered to be the most promising way to describe relativistic viscous fluids.

However, the situation was shown to be more subtle since the relativistic version of Navier-Stokes theory is actually intrinsically **unstable** [8, 9, 10, 11, 12]. The source of such instability is well understood and will be discussed in detail in the next chapter. It comes from the inherent acausal behavior of Navier-Stokes theory [13, 14], which allows signals to propagate with infinite speed. In non-relativistic theories this non-intuitive feature does not give rise to an intrinsic problem and can be ignored. On the other hand, in relativistic systems, where causality is a physical property that is naturally preserved, this feature leads to equations of motion that are intrinsically unstable. Nevertheless, first-order theories are an important initial step to illustrate the basic features of relativistic dissipative fluid dynamics and thus shall be reviewed in this section.

Just like for an ideal fluid, the basic equations of motion for dissipative fluids are given by the conservation laws of (net) particle number and energy-momentum,

$$\partial_\mu N^\mu = 0, \quad (1.48)$$

$$\partial_\mu T^{\mu\nu} = 0. \quad (1.49)$$

However, in the presence of dissipation, the energy-momentum tensor is no longer diagonal and isotropic in the local rest frame. Also, due to diffusion, we expect (net) particle-number flow to appear in the local rest frame of the fluid element. These effects must be taken into account and are introduced in fluid dynamics by adding **dissipative currents**,  $n^\mu$  and  $\tau^{\mu\nu}$ , to the previously derived ideal currents,  $N_{(0)}^\mu$  and  $T_{(0)}^{\mu\nu}$ ,

$$N^\mu = N_{(0)}^\mu + n^\mu = n_0 u^\mu + n^\mu, \quad (1.50)$$

$$T^{\mu\nu} = T_{(0)}^{\mu\nu} + \tau^{\mu\nu} = \varepsilon_0 u^\mu u^\nu - \Delta^{\mu\nu} P_0 + \tau^{\mu\nu}, \quad (1.51)$$

where we indicated equilibrium quantities with a subscript “0”. Here,  $n^\mu$  is the **particle diffusion 4-current**. In order to satisfy angular-momentum conservation,  $\tau^{\mu\nu}$  is defined to be a symmetric tensor,  $\tau^{\mu\nu} = \tau^{\nu\mu}$ . The main problem then becomes to find the dynamical or constitutive equations satisfied by such dissipative currents.

**Exercise 1.5:** Ignoring spin degrees of freedom, the total angular momentum tensor is defined as

$$J^{\lambda,\mu\nu} \equiv x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} . \quad (1.52)$$

Prove that total angular momentum conservation,

$$\partial_\lambda J^{\lambda,\mu\nu} = 0 , \quad (1.53)$$

requires  $T^{\mu\nu}$  (and thus also  $\tau^{\mu\nu}$ ) to be symmetric.

(Hint: use the energy-momentum conservation law (1.49).)

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### 1.3.1 Matching conditions

The introduction of the dissipative currents renders the equilibrium variables ill-defined, since the fluid can no longer be considered to be in local thermodynamic equilibrium. In a viscous fluid, the thermodynamic variables,  $\alpha_0$ ,  $\beta_0$ ,  $s_0$ ,  $P_0$ ,  $\dots$ , can only be defined in terms of a **fictitious equilibrium state** (labeled by the subscript “0”), constructed such that the thermodynamic relations are valid **as if** the fluid were in local thermodynamic equilibrium. The first step to construct such an equilibrium state is to define  $n_0$  and  $\varepsilon_0$  as the **actual (net) particle density**  $n$  and the **actual energy density**  $\varepsilon$  in the local rest frame of the fluid. This is guaranteed by the so-called **Landau matching conditions**,

$$\varepsilon_0 \equiv \varepsilon \equiv u_\nu u_\mu T^{\mu\nu} , \quad (1.54)$$

$$n_0 \equiv n \equiv u_\mu N^\mu . \quad (1.55)$$

The matching conditions (1.54) and (1.55) are enforced by applying the following set of constraints to the dissipative currents,

$$\begin{aligned} u_\nu u_\mu \tau^{\mu\nu} &= 0 , \\ u_\mu n^\mu &= 0 \quad \Longleftrightarrow \quad \Delta^\mu_\lambda n^\lambda = n^\mu . \end{aligned} \quad (1.56)$$

Then, using  $\varepsilon$  and  $n$  we can construct our equilibrium state. The thermodynamic entropy density is determined by the equation of state of the fluid as if in thermodynamic equilibrium,

$$s_0 \equiv s_0(\varepsilon, n) , \quad (1.57)$$

while the remaining thermodynamic variables, e.g., the thermodynamic pressure, temperature, and chemical potential, are defined from the thermodynamic relations derived in Sec. 1.1. The inverse temperature and the ratio of chemical potential over temperature are computed using Eq. (1.13),

$$\beta_0 = \left. \frac{\partial s}{\partial \varepsilon} \right|_n , \quad \alpha_0 = \left. \frac{\partial s}{\partial n} \right|_\varepsilon , \quad (1.58)$$

and the thermodynamic pressure is extracted via Eq. (1.10),

$$P_0 = -\varepsilon + T_0 s_0 + \mu_0 n . \quad (1.59)$$

Note that, in principle, the thermodynamic pressure can also be expressed as a function of other thermodynamic variables, e.g.  $P_0 = P_0(\beta_0, \alpha_0)$ .

It is important to emphasize that, while the energy and (net) particle densities are physically well-defined, all other quantities ( $s_0, P_0, T_0, \mu_0, \dots$ ) are defined only in terms of a fictitious equilibrium state and do not necessarily retain their usual physical meaning. For example, the second law of thermodynamics does not constrain the production of entropy of the fictitious state: it constrains only the production of the actual non-equilibrium entropy of the fluid – a quantity that can be rather nontrivial to construct, as will be discussed later in this chapter.

### 1.3.2 Tensor decomposition of $\tau^{\mu\nu}$

It is convenient to decompose  $\tau^{\mu\nu}$  in terms of its **irreducible components**, i.e., a **scalar**, a **4-vector**, and a **traceless, symmetric second-rank tensor**. This tensor decomposition must respect the matching (or orthogonality) condition satisfied by  $\tau^{\mu\nu}$ , Eq. (1.56). For this purpose, we introduce yet another projection operator: the double-symmetric, traceless rank-4 projection operator orthogonal to  $u^\mu$ ,

$$\Delta_{\alpha\beta}^{\mu\nu} = \frac{1}{2} (\Delta_\alpha^\mu \Delta_\beta^\nu + \Delta_\beta^\mu \Delta_\alpha^\nu) - \frac{1}{\Delta_\lambda^\lambda} \Delta^{\mu\nu} \Delta_{\alpha\beta} , \quad (1.60)$$

which satisfies the following properties,

$$\begin{aligned} \Delta^{(\mu\nu)(\alpha\beta)} &= \Delta^{(\alpha\beta)(\mu\nu)} , \\ \Delta_{\lambda\rho}^{\mu\nu} \Delta_{\alpha\beta}^{\lambda\rho} &= \Delta_{\alpha\beta}^{\mu\nu} , \\ u_\mu \Delta^{\mu\nu\alpha\beta} &= g_{\mu\nu} \Delta^{\mu\nu\alpha\beta} = 0 , \\ \Delta_{\mu\nu}^{\mu\nu} &= 5 . \end{aligned} \quad (1.61)$$

Then, using  $\Delta^{\mu\nu}$  and  $\Delta_{\alpha\beta}^{\mu\nu}$ , the tensor decomposition of  $\tau^{\mu\nu}$  in its irreducible form is implemented as

$$\tau^{\mu\nu} \equiv -\Pi \Delta^{\mu\nu} + 2u^{(\mu} h^{\nu)} + \pi^{\mu\nu} , \quad (1.62)$$

where the parentheses  $()$  denote the symmetrization of all Lorentz indices,  $a^{(\mu\nu)} = (a^{\mu\nu} + a^{\nu\mu})/2$ , and where we defined

$$\Pi \equiv -\frac{1}{3} \Delta_{\alpha\beta}^{\mu\nu} \tau^{\alpha\beta} , \quad h^\mu \equiv \Delta_\alpha^\mu u_\beta \tau^{\alpha\beta} , \quad \pi^{\mu\nu} \equiv \Delta_{\alpha\beta}^{\mu\nu} \tau^{\alpha\beta} . \quad (1.63)$$

The scalar term,  $\Pi$ , is the **bulk-viscous pressure**, the vector term,  $h^\mu$ , is the **energy-diffusion 4-current**, and the second-rank tensor,  $\pi^{\mu\nu}$ , is the **shear-stress tensor**. The properties of the projection operators  $\Delta_\nu^\mu$  and  $\Delta_{\alpha\beta}^{\mu\nu}$  given in Eqs. (1.26) and (1.61) imply that  $h^\mu$  and  $\pi^{\mu\nu}$  satisfy

$$h^\mu = \Delta_\nu^\mu h^\nu \iff u_\mu h^\mu = 0 , \quad (1.64)$$

$$\pi^{\mu\nu} = \pi^{(\mu\nu)} \iff u_\mu \pi^{\mu\nu} = 0 , \quad (1.65)$$

$$\pi_\mu^\mu = 0 , \quad (1.66)$$

where the brackets  $\langle \rangle$  denote the following projection of a second-rank tensor,  $A^{\langle\mu\nu\rangle} \equiv \Delta_{\alpha\beta}^{\mu\nu} A^{\alpha\beta}$ . In summary, the fields  $\Pi$ ,  $h^\mu$ ,  $n^\mu$ , and  $\pi^{\mu\nu}$  are expressed in terms of  $N^\mu$  and  $T^{\mu\nu}$  as

$$\Pi = -P_0 - \frac{1}{3} \Delta_{\mu\nu} T^{\mu\nu} , \quad (1.67)$$

$$h^\mu = u_\alpha \Delta_\beta^\mu T^{\alpha\beta} , \quad (1.68)$$

$$n^\mu = \Delta_\alpha^\mu N^\alpha , \quad (1.69)$$

$$\pi^{\mu\nu} = T^{\langle\mu\nu\rangle} . \quad (1.70)$$

Note that  $T^{\mu\nu}$  is a symmetric second-rank tensor and, thus,  $N^\mu$  and  $T^{\mu\nu}$  have 14 independent components (four from  $N^\mu$  and ten from  $T^{\mu\nu}$ ). In order to satisfy their orthogonality to  $u^\mu$ ,  $n^\mu$  and  $h^\mu$  can only have three independent components each. The shear-stress tensor is symmetric, traceless, and orthogonal to  $u^\mu$  and, therefore, can have only five independent components. Together with  $u^\mu$ ,  $\varepsilon$ ,  $n$ , and  $\Pi$ , which have in total six independent components, we find a total of 17 independent components, three more than expected. This happened because, so far, the velocity field itself has not been specified, being introduced just as a general normalized 4-vector. The definition of the velocity field will provide the three missing constraints that will reduce the number of independent components to the correct value.

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**Exercise 1.6:** Prove Eqs. (1.61), as well as Eqs. (1.63) by explicitly inserting Eq. (1.62) on the right-hand sides.

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### 1.3.3 Definition of the local rest frame and equations of motion

The definition of the velocity field is an important step in deriving fluid dynamics. For ideal fluids, the local rest frame was implicitly defined as the frame in which there is no flow of energy and (net) particle number. Due to the presence of energy and particle diffusion in viscous fluids, this definition is no longer possible. From a mathematical point of view, the velocity can be defined in numerous ways. From the physical perspective, there are, however, two natural choices. The **Landau frame** [2], in which the velocity is defined by the flow of the total energy,

$$u_\mu T^{\mu\nu} = \varepsilon u^\nu , \quad (1.71)$$

and the **Eckart frame** [3, 4], in which the velocity is specified by the flow of (net) particle number,

$$N^\mu = n u^\mu . \quad (1.72)$$

If the system has more than one type of particle (or charge), the Eckart frame must be defined by selecting one of these particle (or charge) types.

Both choices of frame impose different constraints on the dissipative currents introduced in this section. In the Landau frame, the energy diffusion vanishes,

$$h^\mu = 0 , \quad (1.73)$$

while in the Eckart frame the particle diffusion is zero,

$$n^\mu = 0 . \quad (1.74)$$

In other words, in the Landau frame the velocity field is fixed to eliminate any diffusion of energy while in the Eckart frame it is defined to eliminate any diffusion of particles. In this lecture series, we shall always use the Landau frame, Eq. (1.71), and, therefore, the conserved currents take the following simpler form

$$N^\mu = nu^\mu + n^\mu , \quad (1.75)$$

$$T^{\mu\nu} = \varepsilon u^\mu u^\nu - \Delta^{\mu\nu} (P_0 + \Pi) + \pi^{\mu\nu} . \quad (1.76)$$

Note that both the Landau and the Eckart choice of frame reduce the number of independent variables to 14, as advertised at the end of the last section.

As for an ideal fluid, we decompose Eq. (1.49) into a part parallel and another one orthogonal to  $u^\mu$ . As shown in the last section, this is done by projecting and contracting Eq. (1.49) with  $u_\mu$  and  $\Delta^{\mu\nu}$ , i.e., by taking  $u_\alpha \partial_\beta T^{\alpha\beta}$  and  $\Delta_\alpha^\mu \partial_\beta T^{\alpha\beta}$ , respectively. Together with Eqs. (1.48), (1.75), and (1.76), this procedure leads to the equations of motion of the fluid,

$$u_\alpha \partial_\beta T^{\alpha\beta} = \dot{\varepsilon} + (\varepsilon + P_0 + \Pi) \theta - \pi^{\alpha\beta} \sigma_{\alpha\beta} = 0 , \quad (1.77)$$

$$\Delta_\alpha^\mu \partial_\beta T^{\alpha\beta} = (\varepsilon + P_0 + \Pi) \dot{u}^\mu - \nabla^\mu (P_0 + \Pi) + \Delta_\alpha^\mu \partial_\beta \pi^{\alpha\beta} = 0 , \quad (1.78)$$

$$\partial_\mu N^\mu = \dot{n} + n\theta + \partial_\mu n^\mu = 0 , \quad (1.79)$$

where we defined the shear tensor

$$\sigma^{\mu\nu} = \partial^{\langle\mu} u^{\nu\rangle} = \frac{1}{2} (\nabla^\mu u^\nu + \nabla^\nu u^\mu) - \frac{1}{3} \Delta^{\mu\nu} \nabla_\lambda u^\lambda . \quad (1.80)$$

Note that the quantities  $n$ ,  $\varepsilon$ ,  $P_0$ ,  $u^\mu$ ,  $\Pi$ ,  $n^\mu$ , and  $\pi^{\mu\nu}$  introduced in this section were defined from a strict mathematical perspective via the most general tensor decomposition allowed by symmetry. The conservation laws, Eqs. (1.48) and (1.49), the definition of the fictitious equilibrium state, and the definition of the velocity field are also general and valid regardless of whether the system is in the fluid-dynamical regime (i.e., not far from local thermodynamical equilibrium). Thus, by writing down any of the above equations, we have not, by any means, derived fluid dynamics. In order to derive the complete equations of dissipative fluid dynamics, one still has to provide an **additional nine relations** that will close Eqs. (1.77), (1.78), and (1.79). In the end, this corresponds to finding **dynamical or constitutive relations** satisfied by the dissipative currents  $\Pi$ ,  $n^\mu$ , and  $\pi^{\mu\nu}$ . Ideal fluid dynamics, discussed in the previous section, corresponds to a trivial example of this procedure, in which the dissipative currents are simply set to zero.

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**Exercise 1.7:** Prove Eqs. (1.77) – (1.79), extending Exercise 1.3 to dissipative fluids.

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### 1.3.4 Relativistic Navier-Stokes theory

In the presence of dissipative currents the entropy is longer conserved. Deriving the equation for the entropy 4-current is not trivial for a viscous fluid, since *a priori* we do not know the form of this current. For now, let us take the same steps as in the ideal fluid case and see where we arrive. We start by taking Eq. (1.45),

$$\partial_\mu S_{(0)}^\mu = \beta_0 u_\nu \partial_\mu T_{(0)}^{\mu\nu} - \alpha_0 \partial_\mu N_{(0)}^\mu, \quad (1.81)$$

which remains valid in a viscous fluid since, as explained in Sec. 1.3.1, the equilibrium variables and, consequently, the equilibrium currents, were constructed to satisfy thermodynamic relations as if in equilibrium. Now, however, the equilibrium part of the currents are not conserved,  $\partial_\mu N_{(0)}^\mu = -\partial_\mu n^\mu \neq 0$  and  $u_\nu \partial_\mu T_{(0)}^{\mu\nu} = -\Pi\theta + \pi^{\mu\nu}\sigma_{\mu\nu} \neq 0$ , cf. Eqs. (1.77) and (1.79), and in a viscous fluid Eq. (1.81) leads to

$$\partial_\mu S_{(0)}^\mu = \alpha_0 \partial_\mu n^\mu + \beta_0 (-\Pi\theta + \pi^{\mu\nu}\sigma_{\mu\nu}). \quad (1.82)$$

By decomposing the first term on the right-hand side as  $\alpha_0 \partial_\mu n^\mu = \partial_\mu (\alpha_0 n^\mu) - n^\mu \nabla_\mu \alpha_0$ , Eq. (1.82) can be written in a more convenient form,

$$\partial_\mu (S_{(0)}^\mu - \alpha_0 n^\mu) = -n^\mu \nabla_\mu \alpha_0 - \beta_0 \Pi\theta + \beta_0 \pi^{\mu\nu}\sigma_{\mu\nu} \equiv Q. \quad (1.83)$$

It is very tempting to identify the term on the left-hand side of Eq. (1.83) as the 4-divergence of the (off-equilibrium) entropy 4-current

$$S^\mu \equiv S_{(0)}^\mu - \alpha_0 n^\mu = s_0 u^\mu - \alpha_0 n^\mu, \quad (1.84)$$

and the terms on the right-hand side,  $Q$ , as the source terms for entropy production. Note, however, that this is not necessarily the case. Nevertheless, this was the identification proposed by Eckart and by Landau and Lifshitz and we shall consider it here in order to derive **relativistic Navier-Stokes theory**.

Relativistic Navier-Stokes theory is then obtained by applying the second law of thermodynamics to each fluid element, i.e., by requiring that the entropy production obtained in Eq. (1.83) must always be positive semi-definite,  $Q \geq 0$ . The simplest way to satisfy this condition for all possible fluid configurations is to assume that the bulk-viscous pressure, the particle-diffusion 4-current, and the shear-stress tensor are **linearly proportional** to  $\theta$ ,  $\nabla^\mu \alpha_0$ , and  $\sigma^{\mu\nu}$ , respectively,

$$\Pi = -\zeta\theta, \quad (1.85)$$

$$n^\mu = \varkappa \nabla^\mu \alpha_0, \quad (1.86)$$

$$\pi^{\mu\nu} = 2\eta\sigma^{\mu\nu}. \quad (1.87)$$

The proportionality coefficients  $\zeta$ ,  $\varkappa$ , and  $\eta$  are the **coefficients of bulk viscosity**, **particle diffusion**, and **shear viscosity**, respectively. Then, substituting Eqs. (1.85), (1.86), and (1.87) into Eq. (1.83), we see that the entropy production becomes a quadratic function of the dissipative currents

$$Q = \frac{\beta_0}{\zeta} \Pi^2 - \frac{1}{\varkappa} n^\mu n_\mu + \frac{\beta_0}{2\eta} \pi_{\mu\nu} \pi^{\mu\nu}. \quad (1.88)$$

Note that  $n^\mu n_\mu$  is negative ( $n^\mu$  is a space-like vector) while  $\pi_{\mu\nu}\pi^{\mu\nu}$  is positive and, therefore, as long as  $\zeta, \varkappa, \eta \geq 0$ ,  $Q$  is, in fact, always positive semi-definite.

The equations of fluid dynamics are obtained by substituting Eqs. (1.85), (1.86), and (1.87) into the conservation laws, Eqs. (1.77), (1.78), and (1.79),

$$\dot{\varepsilon} = -(\varepsilon + P_0 - \zeta\theta)\theta + 2\eta\sigma_{\alpha\beta}\sigma^{\alpha\beta} , \quad (1.89)$$

$$(\varepsilon + P_0 - \zeta\theta)\dot{u}^\mu = \nabla^\mu P_0 - \nabla^\mu(\zeta\theta) - 2\Delta_\alpha^\mu \partial_\beta(\eta\sigma^{\alpha\beta}) , \quad (1.90)$$

$$\dot{n} = -n\theta - \partial_\mu(\varkappa\nabla^\mu\alpha_0) . \quad (1.91)$$

The above equations are known as the **relativistic Navier-Stokes equations**, as obtained by Landau and Lifshitz [2]. A similar theory was obtained independently by Eckart [4], using a different definition of the local rest frame. In this formulation, the state of a dissipative fluid remains being described by the same variables as in the case of an ideal fluid, i.e., the primary fluid-dynamical variables  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ . The only difference is the existence of **dissipative processes**, corresponding to new forms of particle and energy-momentum transfer, which occur due to **gradients of the primary fluid-dynamical variables**.

As already mentioned, Navier-Stokes theory is acausal and, consequently, unstable. Thus, it is unable to describe any relativistic fluid existing in Nature. The source of the acausality can be understood from the constitutive relations satisfied by the dissipative currents, Eqs. (1.85), (1.86), and (1.87). Such linear relations imply that any inhomogeneity of  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ , will **instantaneously** give rise to a dissipative current. This instantaneous creation of currents from (space-like) gradients of the primary fluid-dynamical variables renders the equations of motion **parabolic**. In a relativistic theory this leads to **instabilities**, as will be discussed in the next chapter.

### 1.3.5 Gradient expansion and Navier-Stokes theory

05/10/2022

Navier-Stokes theory can also be derived (and extended) via the so-called **gradient expansion** [15, 16]. In this framework, the bulk-viscous pressure, the particle-diffusion 4-current, and the shear-stress tensor are assumed to be expressible solely in terms of **powers of gradients of the primary fluid-dynamical variables**  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ . The dissipative currents can then be schematically written as the series

$$\Pi = \lambda_\Pi^{(1)}\mathcal{O}_1 + \lambda_\Pi^{(2)}\mathcal{O}_2 + \dots , \quad (1.92)$$

$$n^\mu = \lambda_n^{(1)}\mathcal{O}_1^\mu + \lambda_n^{(2)}\mathcal{O}_2^\mu + \dots , \quad (1.93)$$

$$\pi^{\mu\nu} = \lambda_\pi^{(1)}\mathcal{O}_1^{\mu\nu} + \lambda_\pi^{(2)}\mathcal{O}_2^{\mu\nu} + \dots , \quad (1.94)$$

where the quantities  $(\mathcal{O}_1, \mathcal{O}_1^\mu, \mathcal{O}_1^{\mu\nu})$  and  $(\mathcal{O}_2, \mathcal{O}_2^\mu, \mathcal{O}_2^{\mu\nu})$  correspond to terms of first and second order in gradients of  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ , respectively, and the dots denote possible higher-order gradient terms.

It is important to remark that when the system exhibits a clear separation between the typical **microscopic** and **macroscopic scales**,  $\lambda$  and  $L$ , respectively, it may possible to truncate the expansion on the right-hand sides of Eqs. (1.92), (1.93), and (1.94). The microscopic scale can, for example, be the **mean free path** for dilute gases or the **inverse**

**temperature** for conformal fluids, while a macroscopic scale is given by **typical time or length scales** over which a certain **primary fluid-dynamical variable** varies, e.g.,  $L \sim [(\partial a)/a]^{-1}$ , with  $a = \alpha_0, \beta_0$ , or  $u^\mu$ . The terms  $\mathcal{O}_1, \mathcal{O}_1^\mu$ , and  $\mathcal{O}_1^{\mu\nu}$  are linearly proportional to a gradient of a macroscopic variable, and thus are of order  $\sim L^{-1}$ . Every additional derivative brings in another inverse power of  $L$  and, thus,  $\mathcal{O}_n, \mathcal{O}_n^\mu, \mathcal{O}_n^{\mu\nu} \sim L^{-n}$ . The microscopic scale  $\lambda$  is contained in the coefficients,  $\lambda_\Pi^{(n)}, \lambda_n^{(n)}$ , and  $\lambda_\pi^{(n)}$ . Up to some overall power of  $\lambda$  (which restores the correct scaling dimension),  $\lambda_\Pi^{(n)}, \lambda_n^{(n)}, \lambda_\pi^{(n)} \sim \lambda^n$ . Therefore, the terms  $(\mathcal{O}_1, \mathcal{O}_1^\mu, \mathcal{O}_1^{\mu\nu})$  and  $(\mathcal{O}_2, \mathcal{O}_2^\mu, \mathcal{O}_2^{\mu\nu})$ , multiplied by their corresponding coefficients in Eqs. (1.92), (1.93), and (1.94) are of order  $\lambda/L$  and  $(\lambda/L)^2$ , respectively. Subsequent terms would be of higher order in this ratio. This is nothing but a series in powers of the so-called **Knudsen number**

$$\text{Kn} \equiv \frac{\lambda}{L} . \quad (1.95)$$

If  $\text{Kn} \ll 1$  and this series converges, the gradient expansion of the dissipative currents can be truncated at a given order and one obtains a closed macroscopic theory for them. **Ideal fluid dynamics** corresponds to the **zeroth-order truncation** of this series, i.e., when no gradient terms are considered. The **first-order truncation** of the gradient expansion is **Navier-Stokes theory**, as will be shown below. Higher-order truncations would lead to the **relativistic Burnett and super-Burnett equations** and so on. We note that the convergence of the gradient expansion is not well established and is a topic that is still being intensely investigated [17].

The first order of the gradient expansion can be obtained by constructing all possible tensors that can be formed from the first-order derivatives of  $\alpha_0, \beta_0$ , and  $u^\mu$ . These can be easily obtained and are

$$\partial_\mu \alpha_0, \partial_\mu \beta_0, \text{ and } \partial_\mu u_\nu . \quad (1.96)$$

Next, using these gradients one has to construct tensors that have the same properties as the dissipative currents. There must be a scalar, such as the bulk-viscous pressure, a 4-vector orthogonal to  $u^\mu$ , such as the particle-diffusion 4-current, and a symmetric, traceless second-rank tensor orthogonal to  $u^\mu$ , such as the shear-stress tensor. The only possibilities are

$$\text{Scalar} : \quad \theta = \nabla_\mu u^\mu , \quad (1.97)$$

$$\text{Vector} : \quad I^\mu \equiv \nabla^\mu \alpha_0, \quad J^\mu \equiv \nabla^\mu \beta_0 , \quad (1.98)$$

$$\text{Tensor} : \quad \sigma^{\mu\nu} \equiv \partial^{(\mu} u^{\nu)} = \frac{1}{2} (\nabla^\mu u^\nu + \nabla^\nu u^\mu) - \frac{1}{3} \Delta^{\mu\nu} \theta . \quad (1.99)$$

Then, the most general first-order terms allowed by symmetry are

$$\mathcal{O}_1 = \theta , \quad (1.100)$$

$$\mathcal{O}_1^\mu = I^\mu + \gamma J^\mu , \quad (1.101)$$

$$\mathcal{O}_1^{\mu\nu} = \sigma^{\mu\nu} , \quad (1.102)$$

where the constant  $\gamma$  is introduced in order to restore the correct dimension.

In order to respect the second law of thermodynamics, discussed in the previous section, the transport coefficient  $\gamma$  must always be zero,  $\gamma = 0$ . This can be seen as follows. One can rewrite the Gibbs-Duhem relation (1.12) with the help of the Euler equation (1.10) in the form

$$d\beta_0 = h_0^{-1} d\alpha_0 - \frac{\beta_0}{\varepsilon + P_0} dP_0, \quad (1.103)$$

where  $h_0 \equiv (\varepsilon + P_0)/n$  is the specific enthalpy. Then, using the hydrodynamic equation (1.78),

$$J^\mu = h_0^{-1} I^\mu - \beta_0 \dot{u}^\mu + \mathcal{O}_2, \quad (1.104)$$

where the second-order terms involve gradients of dissipative currents or products of dissipative currents with gradients of primary fluid-dynamical variables. Employing Eqs. (1.93), (1.101) the first term in the entropy-production equation (1.83) would be  $\sim -\lambda_n^{(1)} [(1 + \gamma h_0^{-1}) I^\mu - \gamma \beta_0 \dot{u}^\mu] I_\mu$ , which is in general no longer positive definite, unless  $\gamma = 0$ .

Therefore, the most general relations satisfied by  $\Pi$ ,  $n^\mu$ , and  $\pi^{\mu\nu}$ , up to first order in  $\text{Kn}$ , are

$$\begin{aligned} \Pi &= \lambda_\Pi^{(1)} \theta, \\ n^\mu &= \lambda_n^{(1)} I^\mu, \\ \pi^{\mu\nu} &= \lambda_\pi^{(1)} \sigma^{\mu\nu}, \end{aligned}$$

which corresponds to the relativistic Navier-Stokes theory [2], with the bulk-viscosity coefficient, the diffusion coefficient, and the shear-viscosity coefficient being identified as  $\zeta \equiv -\lambda_\Pi^{(1)}$ ,  $\varkappa \equiv \lambda_n^{(1)}$ , and  $\eta \equiv \lambda_\pi^{(1)}/2$ , respectively.

In the framework of the gradient expansion, the relativistic Navier-Stokes theory can be extended by including terms of second order in gradients of  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ . In order to do so, one has to obtain all possible terms that can contribute to  $\mathcal{O}_2$ ,  $\mathcal{O}_2^\mu$ , and  $\mathcal{O}_2^{\mu\nu}$ . These are

$$\begin{aligned} \text{Scalar : } & \omega_{\mu\nu} \omega^{\mu\nu}, \sigma_{\mu\nu} \sigma^{\mu\nu}, \theta^2, I_\mu I^\mu, J_\mu J^\mu, I_\mu J^\mu, \nabla_\mu I^\mu, \nabla_\mu J^\mu, \\ \text{Vector : } & \sigma^{\mu\nu} I_\nu, \sigma^{\mu\nu} J_\nu, I^\mu \theta, J^\mu \theta, \omega^{\mu\nu} I_\nu, \omega^{\mu\nu} J_\nu, \Delta_\alpha^\mu \partial_\nu \sigma^{\alpha\nu}, \nabla^\mu \theta, \\ \text{Tensor : } & \omega_\lambda^{\langle\mu} \omega^{\nu\rangle\lambda}, \theta \sigma^{\mu\nu}, \sigma^{\lambda\langle\mu} \sigma_\lambda^{\nu\rangle}, \sigma_\lambda^{\langle\mu} \omega^{\nu\rangle\lambda}, I^{\langle\mu} I^{\nu\rangle}, J^{\langle\mu} J^{\nu\rangle}, \\ & I^{\langle\mu} J^{\nu\rangle}, \nabla^{\langle\mu} I^{\nu\rangle}, \nabla^{\langle\mu} J^{\nu\rangle}, \end{aligned} \quad (1.105)$$

where we introduced the fluid **vorticity tensor**,

$$\omega^{\mu\nu} \equiv \frac{1}{2} (\nabla^\mu u^\nu - \nabla^\nu u^\mu). \quad (1.106)$$

Then, the most general second-order terms allowed by symmetry are

$$\begin{aligned} \lambda_\Pi^{(2)} \mathcal{O}_2 &= \zeta_1 \omega_{\mu\nu} \omega^{\mu\nu} + \zeta_2 \sigma_{\mu\nu} \sigma^{\mu\nu} + \zeta_3 \theta^2 + \zeta_4 I_\mu I^\mu + \zeta_5 J_\mu J^\mu + \zeta_6 I_\mu J^\mu \\ &+ \zeta_7 \nabla_\mu I^\mu + \zeta_8 \nabla_\mu J^\mu, \end{aligned} \quad (1.107)$$

$$\begin{aligned} \lambda_n^{(2)} \mathcal{O}_2^\mu &= \varkappa_1 \sigma^{\mu\nu} I_\nu + \varkappa_2 \sigma^{\mu\nu} J_\nu + \varkappa_3 I^\mu \theta + \varkappa_4 J^\mu \theta + \varkappa_5 \omega^{\mu\nu} I_\nu + \varkappa_6 \omega^{\mu\nu} J_\nu \\ &+ \varkappa_7 \Delta_\alpha^\mu \partial_\nu \sigma^{\alpha\nu} + \varkappa_8 \nabla^\mu \theta, \end{aligned} \quad (1.108)$$

$$\begin{aligned} \lambda_\pi^{(2)} \mathcal{O}_2^{\mu\nu} &= \eta_1 \omega_\lambda^{\langle\mu} \omega^{\nu\rangle\lambda} + \eta_2 \theta \sigma^{\mu\nu} + \eta_3 \sigma^{\lambda\langle\mu} \sigma_\lambda^{\nu\rangle} + \eta_4 \sigma_\lambda^{\langle\mu} \omega^{\nu\rangle\lambda} + \eta_5 I^{\langle\mu} I^{\nu\rangle} \\ &+ \eta_6 J^{\langle\mu} J^{\nu\rangle} + \eta_7 I^{\langle\mu} J^{\nu\rangle} + \eta_8 \nabla^{\langle\mu} I^{\nu\rangle} + \eta_9 \nabla^{\langle\mu} J^{\nu\rangle}, \end{aligned} \quad (1.109)$$

where we introduced **additional transport coefficients**,  $\zeta_i$ ,  $\varkappa_i$ , and  $\eta_i$ . By including the above terms in the expressions for the dissipative currents, Eqs. (1.92), (1.93), and (1.94), we obtain the **relativistic Burnett equations** [16],

$$\begin{aligned} \Pi &= -\zeta\theta + \zeta_1\omega_{\mu\nu}\omega^{\mu\nu} + \zeta_2\sigma_{\mu\nu}\sigma^{\mu\nu} + \zeta_3\theta^2 + \zeta_4I_\mu I^\mu + \zeta_5J_\mu J^\mu + \zeta_6I_\mu J^\mu \\ &\quad + \zeta_7\nabla_\mu I^\mu + \zeta_8\nabla_\mu J^\mu, \end{aligned} \quad (1.110)$$

$$\begin{aligned} n^\mu &= \varkappa I^\mu + \varkappa_1\sigma^{\mu\nu}I_\nu + \varkappa_2\sigma^{\mu\nu}J_\nu + \varkappa_3I^\mu\theta + \varkappa_4J^\mu\theta + \varkappa_5\omega^{\mu\nu}I_\nu + \varkappa_6\omega^{\mu\nu}J_\nu \\ &\quad + \varkappa_7\Delta_\alpha^\mu\partial_\nu\sigma^{\alpha\nu} + \varkappa_8\nabla^\mu\theta, \end{aligned} \quad (1.111)$$

$$\begin{aligned} \pi^{\mu\nu} &= 2\eta\sigma^{\mu\nu} + \eta_1\omega_\lambda^{\langle\mu}\omega^{\nu\rangle\lambda} + \eta_2\theta\sigma^{\mu\nu} + \eta_3\sigma^{\lambda\langle\mu}\sigma_\lambda^{\nu\rangle} + \eta_4\sigma_\lambda^{\langle\mu}\omega^{\nu\rangle\lambda} + \eta_5I^{\langle\mu}I^{\nu\rangle} \\ &\quad + \eta_6J^{\langle\mu}J^{\nu\rangle} + \eta_7I^{\langle\mu}J^{\nu\rangle} + \eta_8\nabla^{\langle\mu}I^{\nu\rangle} + \eta_9\nabla^{\langle\mu}J^{\nu\rangle}. \end{aligned} \quad (1.112)$$

In this section, we showed how to extend and derive Navier-Stokes theory via the gradient expansion. Note, however, that these extensions remain **acausal** and **unstable** and have no practical purpose. As a matter of fact, the Burnett equations are unstable even in the non-relativistic regime [18].

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**Exercise 1.8:** Check the validity of the thermodynamic relation (1.103).

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## 1.4 Causal fluid dynamics

Many theories have been developed to incorporate dissipative effects in fluid dynamics preserving causality: **Grad-Israel-Stewart theory** [5, 6, 7, 19], **divergence-type theories** [20, 21, 22], **extended irreversible thermodynamics** [23, 24, 25, 26, 27], **Carter's theory** [28], **Öttinger-Grmela theory** [29], among others [30, 31]. In Sec. 1.5 we shall briefly review Israel and Stewart's approach.

However, before explaining Israel-Stewart theory, it is useful to discuss a more *ad hoc* approach to render Navier-Stokes theory causal and stable [32]. For the sake of simplicity, we first illustrate this method using the simple example of a transport equation that shares the same problems as Navier-Stokes theory: the **heat-conduction equation**.

### 1.4.1 Diffusion equation and acausality in heat conduction

The fundamental problem of the diffusion equation comes from the fact that it is a **parabolic equation**, allowing for signals that can propagate with infinite speed [25, 26, 27, 33, 34, 35]. In the non-relativistic case, this problem was first addressed by Cattaneo [36, 37] and applied to heat conduction. Cattaneo argued that the problem of acausal propagation in the diffusion equation

$$\partial_t A = D\nabla^2 A, \quad (1.113)$$

can be corrected by introducing a term with a **second-order time derivative**, i.e.,

$$\tau_R\partial_t^2 A + \partial_t A = D\nabla^2 A, \quad (1.114)$$

thereby converting a parabolic equation to a **hyperbolic one**. Above, we introduced the **diffusion** and the **relaxation-time coefficients**,  $D$  and  $\tau_R$ , respectively. Equation (1.114) is often referred to as **telegraph equation** and, for suitable choices of  $D$  and  $\tau_R$ , can lead to **causal signal propagation**. As a matter of fact, the maximum propagation speed of signals in this theory can be proven to be [38],

$$v_{\max} = \sqrt{D/\tau_R} . \quad (1.115)$$

Therefore, as long as  $D/\tau_R \leq 1$ , the telegraph equation is causal. On the other hand, in the limit  $\tau_R \rightarrow 0$ , in which the diffusion equation is recovered, the propagation speed diverges and the theory becomes acausal.

Next, we consider the heat-conduction problem as an example to understand the physical origin of the telegraph equation. The diffusion equation used to describe heat conduction is constructed from two basic features. One is the **energy-balance equation**,

$$\rho c \partial_t T + \nabla \cdot \mathbf{J} = 0 , \quad (1.116)$$

where  $T$  is the **temperature**,  $\mathbf{J}$  is the **heat flux**,  $\rho$  is the **mass density**, and  $c$  is the **specific heat capacity** of the material in question. The other ingredient is **Fourier's law**,

$$\mathbf{J} = -\varpi \nabla T , \quad (1.117)$$

where  $\varpi$  is the **heat conductivity**. Then, Eqs. (1.116) and (1.117) can be combined to eliminate  $\mathbf{J}$  from the differential equation (1.116) and describe heat conduction via the following **diffusion equation**,

$$\partial_t T = \frac{\varpi}{\rho c} \nabla^2 T , \quad (1.118)$$

where, for the sake of simplicity, we assumed that  $\rho$  and  $\varpi$  are spatially constant in the material.

The source of acausality cannot be the energy-balance equation, which is the direct consequence of a conservation law. Then, it must be Fourier's law. Heat conduction should always be induced by an inhomogeneous temperature profile. However, Fourier's law is only an approximation of this process. It does not contain any inertial effects and, consequently, certain perturbations in the temperature distribution are felt **instantaneously** at every point of the system, i.e., according to Eq. (1.118) **every point** in the material heats at the **same time**, no matter how distant from the heat source.

Therefore, even though very successful in non-relativistic applications, Fourier's law cannot be employed in the relativistic limit: in relativistic theories, causality is a physical principle that is naturally preserved by the Lorentz transformations and dictates that the propagation of any current must happen in a nonzero interval of time. This feature can be included in the description of heat conduction via **linear response theory**, which prescribes a more general expression for  $\mathbf{J}$  [39],

$$\mathbf{J}(t) = - \int^t ds G(t-s) \nabla T(s) , \quad (1.119)$$

where  $G(t)$  is the corresponding **retarded Green function**. The Green function includes the **microscopic time scales** that describe the creation of the heat flux from temperature

gradients. In non-relativistic systems in which the separation between the microscopic and macroscopic time scales of the system are sufficiently large, the time dependence of the Green function can be approximated by a Dirac delta function,  $G(t) \sim \delta(t)$ , and we recover Fourier's law. However, when the microscopic and macroscopic time scales are not separated by orders of magnitude, the **transient dynamics** of the heat flux must be explicitly described and the time dependence of  $G(t)$  must be taken into account. In relativistic systems, such microscopic time scales must always be considered in order to preserve causality. When deriving fluid dynamics in the next chapters, we shall address this topic within a more formal framework.

The simplest choice for  $G(t)$  is the exponential *ansatz* (this choice can actually be derived in the framework of kinetic theory) [40],

$$G(t) = \frac{\varpi}{\tau_R} e^{-t/\tau_R} , \quad (1.120)$$

where  $\tau_R$  is the **heat-flux relaxation time**. Then, by taking the time derivative of Eq. (1.119) and substituting this *ansatz*, we obtain the following equation of motion for  $\mathbf{J}$ ,

$$\tau_R \partial_t \mathbf{J}(t) + \mathbf{J}(t) = -\varpi \nabla T(t) . \quad (1.121)$$

This equation is often referred to as **Maxwell-Cattaneo equation**. In this theory, the heat flux is not created instantaneously from temperature inhomogeneities. For example, when  $\nabla T = 0$ ,

$$\tau_R \partial_t \mathbf{J}(t) + \mathbf{J}(t) = 0 , \quad (1.122)$$

the heat flux does not vanish instantaneously, as happened in Fourier's law, but **relaxes exponentially to zero** on times scales given by the heat-flux relaxation time  $\tau_R$ .

By using the energy-balance equation (1.116), we can eliminate  $\mathbf{J}$  from the divergence of the Maxwell-Cattaneo equation (1.121), and obtain the **telegraph equation** for the temperature  $T$ ,

$$\tau_R \partial_t^2 T + \partial_t T = \frac{\varpi}{\rho c} \nabla^2 T . \quad (1.123)$$

Note that, in this case, the diffusion equation becomes the **asymptotic limit** of the telegraph equation, attained only for times much longer than the heat-flux relaxation time (as long as  $\nabla T$  varies slowly in time). As mentioned before, Eq. (1.123) is **causal** as long as

$$\sqrt{\frac{\varpi}{\rho c \tau_R}} \leq 1 . \quad (1.124)$$

### 1.4.2 Transient theory of fluid dynamics

The same idea as explained in the previous section can be applied to render Navier-Stokes theory causal. In Sec. 1.3.4, Navier-Stokes theory was constructed from the conservation laws of (net) particle number, energy, and momentum, and the constitutive relations satisfied by the dissipative currents, Eqs. (1.85), (1.86), and (1.87). The conservation laws come from general physical principles, valid even outside the fluid-dynamical regime, and cannot be modified. Therefore, in order to improve on Navier-Stokes theory, we



must extend the constitutive relations (1.85), (1.86), and (1.87). We have already argued that such relations are the source of causality violations in Navier-Stokes theory since, as happens with Fourier's law, they allow for the **instantaneous creation** of dissipative currents from gradients of  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ . In this section, we show how to correct this unphysical behavior using the same arguments as previously applied to heat conduction: the inclusion of a **time delay** in the creation of the dissipative currents from gradients of the primary fluid-dynamical variables.

In the heat-conduction case, inertial effects on the creation of heat flux from temperature inhomogeneities were included by introducing a term with a **first-order time derivative** of  $\mathbf{J}$  in Fourier's law, giving rise to a **causal transport equation** for heat conduction, Eq. (1.122). Similarly, relaxation effects can be introduced in Navier-Stokes theory by adding a term of **first order in the comoving derivative of each dissipative current** to the constitutive relation satisfied by this current, i.e.,  $\dot{\Pi}$ ,  $\Delta_\alpha^\mu \dot{n}^\alpha$ , and  $\Delta_{\alpha\beta}^{\mu\nu} \dot{\pi}^{\alpha\beta}$ . Then, instead of Navier-Stokes theory, we obtain the following transport equations for  $\Pi$ ,  $n^\mu$ , and  $\pi^{\mu\nu}$ ,

$$\tau_\Pi \dot{\Pi} + \Pi = -\zeta\theta + \dots, \quad (1.125)$$

$$\tau_n \Delta_\alpha^\mu \dot{n}^\alpha + n^\mu = \varkappa \nabla^\mu \alpha_0 + \dots, \quad (1.126)$$

$$\tau_\pi \Delta_{\alpha\beta}^{\mu\nu} \dot{\pi}^{\alpha\beta} + \pi^{\mu\nu} = 2\eta \sigma^{\mu\nu} + \dots, \quad (1.127)$$

where we introduced the **bulk relaxation time**,  $\tau_\Pi$ , the **diffusion relaxation time**,  $\tau_n$ , and the **shear relaxation time**,  $\tau_\pi$ . The dots denote possible nonlinear terms involving the fluid-dynamical quantities and their gradients.

In this formulation, the fluid-dynamical dissipative currents appear as **independent dynamical variables**, which **relax** to the values of Navier-Stokes theory on **characteristic time scales** given by the relaxation times  $\tau_\Pi$ ,  $\tau_n$ , and  $\tau_\pi$ . Thus, unlike for the gradient expansion, the dissipative currents in this theory do not have to be zero in the absence of gradients. Instead, they **decay to zero** on the time scales given by the relaxation times. This type of formalism was referred to as **transient theory of fluid dynamics** by Israel and Stewart [5], since it describes this relaxation (or transient) process of each dissipative current towards its respective (asymptotic) Navier-Stokes value.

One of the features of transient fluid dynamics is that it reduces to Navier-Stokes theory in the limit of vanishing relaxation times. In other words, in Navier-Stokes theory the dissipative currents relax instantaneously to their Navier-Stokes values (also referred to as fluid-dynamical forces), which leads to a violation of causality. In many non-relativistic fluids, the relaxation times are very short, and such transient dynamics can be neglected. In this case, the dissipative currents can actually be well approximated by their Navier-Stokes solution. Nevertheless, in the relativistic case, this approximation is not possible since it will render the equations of motion parabolic and unstable.

For fluid dynamics to be causal it is therefore **necessary** that the relaxation times assume a non-zero value, but this is **not sufficient**. As will be shown in the next chapter (see also Refs. [13, 14]) causality imposes a **stronger** constraint for transient theories: the ratio of the relaxation times to their respective viscosity coefficients must **exceed certain values**. It will be also shown that, for relativistic fluids, **causality implies stability** of the fluid-dynamical equations.



Equations (1.125) – (1.127) correspond to the type of relativistic fluid-dynamical theory that we expect to derive from microscopic theory, including all the nonlinear and higher-order terms that might appear on the right-hand side. This will be shown explicitly in the later chapters of this lecture series.

## 1.5 Transient thermodynamics and Israel-Stewart theory

It is also possible to derive causal fluid-dynamical equations, with the same structure as Eqs. (1.125) – (1.127), from the **second law of thermodynamics**. In this section, we review this derivation as first proposed by Israel and Stewart [5, 7]. The main idea of their approach is to apply the second law of thermodynamics to a more general expression of the non-equilibrium entropy 4-current. In equilibrium, the entropy 4-current was expressed exactly in terms of the primary fluid-dynamical variables,  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ . When the fluid deviates from equilibrium, the situation becomes more complicated. Strictly speaking, the entropy 4-current should depend on a very large number of independent dynamical variables (for a dilute gas, these correspond to all the moments of the Boltzmann equation) that are needed in order to characterize the complicated state of a non-equilibrium system. However, it is reasonable to assume that, as the system approaches equilibrium, the number of dynamical variables needed to describe the state of the fluid gradually decreases, until it reaches the variables required by the equilibrium state,  $\alpha_0$ ,  $\beta_0$ , and  $u^\mu$ .

In the previous section, we showed that, in order to render the fluid-dynamical equations causal, the dissipative currents must be promoted to **independent dynamical variables**. Therefore, we expect that a more realistic description of the entropy 4-current can be obtained by considering it to be a function not only of the primary fluid-dynamical variables, but also of the dissipative currents,

$$S^\mu = S^\mu(\alpha_0, \beta_0, u^\mu, \Pi, n^\mu, \pi^{\mu\nu}) . \quad (1.128)$$

Mathematically, it is further assumed that the entropy 4-current has the following properties: (i) it is **additive**; (ii) it is a **convex function** of the equilibrium variables **and** the dissipative currents; and (iii) the corresponding **entropy production is locally positive**. We remark that, while here these properties enter as hypothesis, they can be rigorously derived in the framework of kinetic theory [5, 6, 7].

Then, the entropy 4-current can be expanded in terms of **powers of the dissipative currents** around a (fictitious) equilibrium state [5, 7],

$$S^\mu = S_{(0)}^\mu - \alpha_0 n^\mu + Q^\mu + \mathcal{O}_3 , \quad (1.129)$$

where  $\mathcal{O}_3$  denotes terms of third order or higher in the dissipative currents and

$$Q^\mu \equiv -\frac{1}{2} u^\mu (\delta_0 \Pi^2 - \delta_1 n_\alpha n^\alpha + \delta_2 \pi_{\alpha\beta} \pi^{\alpha\beta}) - \gamma_0 \Pi n^\mu - \gamma_1 \pi_\nu^\mu n^\nu \quad (1.130)$$

is of second order,  $Q^\mu \sim \mathcal{O}_2$ . The expansion coefficients,  $\delta_0$ ,  $\delta_1$ ,  $\delta_2$ ,  $\gamma_0$ , and  $\gamma_1$ , are complicated functions of the temperature and chemical potential of the (fictitious) equilibrium state and can only be obtained by matching this expansion with the underlying

microscopic theory. Note that the entropy 4-current used to derive relativistic Navier-Stokes theory is recovered by taking  $Q^\mu = 0$ . It is important to remember that  $Q^\mu$  is not orthogonal to the fluid 4-velocity and, consequently,

$$s \equiv u_\mu S^\mu = s_0 + u_\mu Q^\mu \neq s_0 . \quad (1.131)$$

That is, the non-equilibrium entropy density in the local rest frame,  $s$ , does **not** correspond to the entropy density computed using the (fictitious) equilibrium state,  $s_0(\varepsilon, n)$ .

The existence of second-order contributions to the entropy 4-current will affect all previous conclusions drawn from the second law of thermodynamics, which can then be understood to be valid only up to **first order** in the dissipative currents (hence the name first-order theory). Next, we re-calculate the entropy production using the more general entropy 4-current introduced in Eq. (1.129),

$$\partial_\mu S^\mu = \beta_0 u_\nu \partial_\mu T_{(0)}^{\mu\nu} - \alpha_0 \partial_\mu N_{(0)}^\mu - \partial_\mu (\alpha_0 n^\mu) + \partial_\mu Q^\mu , \quad (1.132)$$

where we employed Eq. (1.45). The conservation laws (1.77) and (1.79) lead to the following result,

$$\partial_\mu S^\mu = -\beta_0 \Pi \dot{\theta} + \beta_0 \pi^{\mu\nu} \sigma_{\mu\nu} - n^\mu \nabla_\mu \alpha_0 + \partial_\mu Q^\mu . \quad (1.133)$$

Using Eq. (1.130), we can derive all terms originating from  $\partial_\mu Q^\mu$ ,

$$\begin{aligned} \partial_\mu Q^\mu = & -\delta_0 \Pi \dot{\Pi} + \delta_1 n_\mu \dot{n}^\mu - \delta_2 \pi_{\mu\nu} \dot{\pi}^{\mu\nu} - \frac{1}{2} \left( \Pi^2 \dot{\delta}_0 - n_\mu n^\mu \dot{\delta}_1 + \pi_{\mu\nu} \pi^{\mu\nu} \dot{\delta}_2 \right) \\ & - \frac{1}{2} \left( \delta_0 \Pi^2 - \delta_1 n_\mu n^\mu + \delta_2 \pi_{\mu\nu} \pi^{\mu\nu} \right) \theta - \gamma_0 \Pi \partial_\mu n^\mu - \gamma_0 n^\mu \nabla_\mu \Pi - \Pi n^\mu \nabla_\mu \gamma_0 \\ & - \gamma_1 \pi_{\mu\nu} \nabla^{(\mu} n^{\nu)} - \pi_{\mu\nu} n^{(\mu} \nabla^{\nu)} \gamma_1 - \gamma_1 n_\nu \partial_\mu \pi^{\mu\nu} . \end{aligned} \quad (1.134)$$

Then, substituting Eq. (1.134) into Eq. (1.133), we obtain the more general entropy-production equation

$$\begin{aligned} \partial_\mu S^\mu = & \beta_0 \Pi \left( -\theta - \frac{\delta_0}{\beta_0} \dot{\Pi} - \frac{1}{2\beta_0} \Pi \dot{\delta}_0 - \frac{1}{2\beta_0} \delta_0 \Pi \theta - \frac{\gamma_0}{\beta_0} \partial_\mu n^\mu - \frac{1-r}{\beta_0} n^\mu \nabla_\mu \gamma_0 \right) \\ & + n_\mu \left( -\nabla^\mu \alpha_0 + \delta_1 \Delta_\alpha^\mu \dot{n}^\alpha + \frac{n^\mu}{2} \dot{\delta}_1 + \frac{\delta_1}{2} n^\mu \theta - \gamma_0 \nabla^\mu \Pi - r \Pi \nabla^\mu \gamma_0 - \gamma_1 \partial_\nu \pi^{\mu\nu} - y \pi^{\mu\nu} \nabla_\nu \gamma_1 \right) \\ & + \beta_0 \pi_{\mu\nu} \left( \sigma^{\mu\nu} - \frac{\delta_2}{\beta_0} \Delta_{\alpha\beta}^{\mu\nu} \dot{\pi}^{\alpha\beta} - \frac{1}{2\beta_0} \pi^{\mu\nu} \dot{\delta}_2 - \frac{1}{2\beta_0} \delta_2 \pi^{\mu\nu} \theta - \frac{\gamma_1}{\beta_0} \nabla^{(\mu} n^{\nu)} - \frac{1-y}{\beta_0} n^{(\mu} \nabla^{\nu)} \gamma_1 \right) , \end{aligned} \quad (1.135)$$

where  $r, y$  are arbitrary constants.

As argued before, the only way to explicitly satisfy the second law of thermodynamics is to assure that the entropy production is a **positive semi-definite quadratic function of the dissipative currents**, i.e.,

$$\partial_\mu S^\mu \equiv \beta_0 \varpi_\Pi \Pi^2 - \varpi_n n_\mu n^\mu + \beta_0 \varpi_\pi \pi_{\mu\nu} \pi^{\mu\nu} , \quad (1.136)$$

where  $\varpi_\Pi$ ,  $\varpi_n$ ,  $\varpi_\pi \geq 0$ . This further implies that the dissipative currents must satisfy the following **dynamical** equations

$$\begin{aligned}
 \frac{\delta_0}{\beta_0} \dot{\Pi} + \varpi_\Pi \Pi &= -\theta - \frac{1}{2\beta_0} \Pi \dot{\delta}_0 - \frac{1}{2\beta_0} \delta_0 \Pi \theta - \frac{\gamma_0}{\beta_0} \partial_\mu n^\mu - \frac{1-r}{\beta_0} n^\mu \nabla_\mu \gamma_0 , \\
 \delta_1 \Delta_\alpha^\mu \dot{n}^\alpha + \varpi_n n^\mu &= \nabla^\mu \alpha_0 - \frac{1}{2} n^\mu \dot{\delta}_1 - \frac{\delta_1}{2} n^\mu \theta + \gamma_0 \nabla^\mu \Pi + r \Pi \nabla^\mu \gamma_0 + \gamma_1 \partial_\nu \pi^{\mu\nu} \\
 &\quad + y \pi^{\mu\nu} \nabla_\nu \gamma_1 , \\
 \frac{\delta_2}{\beta_0} \Delta_{\alpha\beta}^{\mu\nu} \dot{\pi}^{\alpha\beta} + \varpi_\pi \pi^{\mu\nu} &= \sigma^{\mu\nu} - \frac{1}{2\beta_0} \pi^{\mu\nu} \dot{\delta}_2 - \frac{1}{2\beta_0} \delta_2 \pi^{\mu\nu} \theta - \frac{\gamma_1}{\beta_0} \nabla^{(\mu} n^{\nu)} - \frac{1-y}{\beta_0} n^{(\mu} \nabla^{\nu)} \gamma_1 ,
 \end{aligned} \tag{1.137}$$

which are relaxation-type equations, similar to those conjectured in the last section, i.e., Eqs. (1.125) – (1.127). By comparison with those equations, we find  $\varpi_\Pi$ ,  $\varpi_n$ , and  $\varpi_\pi$  to be related to the viscosity and diffusion coefficients,

$$\varpi_\Pi = \frac{1}{\zeta} , \quad \varpi_n = \frac{1}{\varkappa} , \quad \varpi_\pi = \frac{1}{2\eta} , \tag{1.138}$$

and identify the relaxation times as

$$\tau_\Pi = \zeta \frac{\delta_0}{\beta_0} , \quad \tau_n = \varkappa \delta_1 , \quad \tau_\pi = 2\eta \frac{\delta_2}{\beta_0} . \tag{1.139}$$

Since the relaxation times must be positive, the expansion coefficients  $\delta_0$ ,  $\delta_1$ , and  $\delta_2$  must all be larger than zero. Furthermore, we found some of the nonlinear terms that may appear as source terms in the transient equations of motion for the dissipative currents. We shall see in the next chapters, when we derive the equations of fluid dynamics from microscopic theory, that there are still other nonlinear terms that are missing in this type of derivation. A derivation from microscopic theory also allows to uniquely fix the, as of yet, arbitrary constants  $r, y$ .

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## 2 Linear Stability and Causality

It was emphasized in the previous chapter that relativistic generalizations of Navier-Stokes theory, derived by Landau and Lifshitz [1] and, independently, by Eckart [2], are ill-defined, containing unphysical instabilities when perturbed around an arbitrary global-equilibrium state [3, 4, 5]. Such instabilities are intrinsically related to the acausal nature of Navier-Stokes theory, which allows for perturbations that propagate with an infinite speed. These fundamental problems prohibit the application of relativistic Navier-Stokes theory to describe any practical fluid-dynamical problem, may it be in the description of neutron-star mergers or in the description of the quark-gluon plasma produced in heavy-ion collisions.

In order to address these fundamental issues, Israel and Stewart constructed stable and causal theories of relativistic fluid dynamics, following the procedure initially developed by H. Grad [6] for non-relativistic systems. Israel and Stewart performed this task in two distinct ways [7]. The first is a phenomenological derivation, based on the second law of thermodynamics, which was discussed in detail in Chapter 1. The second is a microscopic derivation starting from the relativistic Boltzmann equation, which will be discussed thoroughly in one of the following chapters. Similar theories have been widely developed in the past decades [8, 9, 10, 11, 12, 13, 14, 15, 16], but all carry the same fundamental aspect: in contrast to Navier-Stokes theory, such causal theories of fluid dynamics include in their description the transient dynamics of the non-conserved dissipative currents. For this reason, they were initially named by Israel and Stewart as **transient fluid dynamics** (nowadays, they are often referred to as **second-order theories**).

However, it is important to remark that the theory formulated by Israel and Stewart is **not guaranteed** to be causal and stable. As was first shown by Hiscock, Lindblom, and later by Olson, such transient theories of fluid dynamics are **only causal and stable** if their **transport coefficients satisfy certain conditions** [17, 18, 19]. Such conclusions were obtained by analysing the properties of the theory in the linear regime and by imposing that the perturbations around a global-equilibrium state are stable and propagate subluminally. Such stability analyses were more recently revisited in Ref. [4], including only the effects of bulk viscosity, and in Ref. [5], which included the effects of both shear and bulk viscosity. In both these papers, constraints for the shear and bulk relaxation times were explicitly derived. Nowadays, the causality of relativistic fluid-dynamical theories has been investigated even in the nonlinear regime [20, 21] (including the effects of shear and bulk viscosity), where more general inequalities required to ensure causality were derived. In the latter case, the inequalities constrain not only the transport coefficients, but also the values of the dissipative currents (in the linear regime, the inequalities derived in Ref. [21] reduce to those derived in Refs. [4, 5]). Such constraints are relevant for, e.g., fluid-dynamical applications to heavy-ion collisions, since the transport coefficients of QCD matter are not precisely known (often, they are completely unknown) and

such fundamental constraints on transport coefficients (and the values of the dissipative currents) can be extremely useful.

In this chapter we perform a **linear stability analysis** of **relativistic Navier-Stokes theory** and of **Israel-Stewart theory** around a general global-equilibrium state. We then demonstrate explicitly the results described above. First, we prove that the general global-equilibrium state in relativistic Navier-Stokes theory is unstable, due to the appearance of so-called non-hydrodynamic modes that grow exponentially on microscopic time scales. We then demonstrate that the Israel-Stewart theory remains stable, as long as the transport coefficients of the linearized theory satisfy certain constraints.

This chapter is organized as follows. In Sec. 2.1, we first write the fluid-dynamical equations by approximating them to linear order in perturbations around a global-equilibrium background. In Sec. 2.2 we transform these equations into Fourier space, with the purpose of determining the dispersion relations satisfied by the perturbations. Section 2.3 discusses these dispersion relations for ideal fluid dynamics, where one just has stable sound modes. In Sec. 2.4 we investigate the causality and stability of the linearized fluid-dynamical equations in the Navier-Stokes limit. Although these equations appear to be stable in a static background, they will become **unstable when the perturbations are performed on a moving background**. Section 2.5 then discusses stability and causality of transient theories of relativistic fluid dynamics. **Both in a static as well as in a moving background these theories are causal and stable**, provided that certain **asymptotic causality conditions are fulfilled**. A summary of our results concludes this chapter in Sec. 2.6.

## 2.1 Fluid-dynamical equations linearized around global equilibrium

In this section, we linearize the fluid-dynamical equations described in the previous chapter around a global-equilibrium state. In their linearized form, the equations simplify considerably and some of their properties can be studied systematically. In particular, our goal is to discuss the stability of relativistic fluids and to verify under which circumstances acausal modes appear in the linearized theory. Here, we consider small fluid-dynamical perturbations around a global-equilibrium state, with inverse temperature  $\beta_0 \equiv 1/T_0$ , thermal potential  $\alpha_0 = \beta_0 \mu_0$ , where  $\mu_0$  is the chemical potential, and a velocity  $u_0^\mu$ , satisfying the normalization condition,  $u_{0\mu} u_0^\mu = 1$ . The pressure is obtained using an arbitrary equation of state,  $P_0 \equiv P(\beta_0, \alpha_0)$ . Naturally, in this state the dissipative currents appearing in the net-charge 4-current,  $N^\mu$ , and the energy-momentum tensor,  $T^{\mu\nu}$ , all vanish:  $\Pi_0 = \pi_0^{\mu\nu} = n_0^\mu = h_0^\mu = 0$ .

In Chapter 1, we derived the equations of motion of a relativistic fluid and discussed two possible definitions of the fluid 4-velocity: one proposed by Landau [1] and another by Eckart [2]. We note that the prescription chosen for the velocity field can affect some aspects of the linear stability analyses [19]. Here, we perform our analysis using Landau's prescription for the velocity field, i.e.,  $T^{\mu\nu} u_\nu = \varepsilon u^\mu$ . In this case, the continuity equations

related to energy, momentum, and net-charge conservation are given by

$$D\varepsilon + (\varepsilon + P + \Pi)\theta - \pi^{\alpha\beta}\sigma_{\alpha\beta} = 0, \quad (2.1)$$

$$(\varepsilon + P + \Pi)Du^\mu - \nabla^\mu(P + \Pi) - \pi^{\mu\beta}Du_\beta + \Delta_\alpha^\mu\nabla_\beta\pi^{\alpha\beta} = 0, \quad (2.2)$$

$$Dn + n\theta - n^\mu Du_\mu + \nabla_\mu n^\mu = 0. \quad (2.3)$$

Here,  $\varepsilon$  is the energy density,  $n$  is the net-charge density,  $D \equiv u^\mu\partial_\mu$  is the comoving derivative,  $\theta \equiv \partial_\mu u^\mu$  is the expansion scalar,  $\sigma^{\mu\nu} \equiv \partial^{(\mu}u^{\nu)}$  is the shear tensor,  $\Delta_\nu^\mu = g_\nu^\mu - u^\mu u_\nu$  is the projection operator onto the 3-space orthogonal to  $u^\mu$ , and  $\nabla^\mu \equiv \partial^{(\mu)}$ ,  $A^{(\mu)} \equiv \Delta_\nu^\mu A^\nu$ ,  $A^{(\mu\nu)} \equiv \Delta_{\alpha\beta}^{\mu\nu} A^{\alpha\beta}$ , with  $\Delta^{\mu\nu\alpha\beta} = \frac{1}{2}(\Delta^{\mu\alpha}\Delta^{\nu\beta} + \Delta^{\mu\beta}\Delta^{\nu\alpha}) - \frac{1}{3}\Delta^{\mu\nu}\Delta^{\alpha\beta}$ .

As discussed in Sec. 1.4.2, in transient theories of fluid dynamics the dissipative currents are determined from **dynamical evolution equations**, and not by **static constitutive relations**. These equations of motion were already derived in the previous chapter and are given by

$$\tau_\Pi D\Pi + \Pi = -\zeta\theta + \dots, \quad (2.4)$$

$$\tau_n \Delta_\alpha^\mu Dn^\alpha + n^\mu = \varkappa \nabla^\mu \alpha + \dots, \quad (2.5)$$

$$\tau_\pi \Delta_{\alpha\beta}^{\mu\nu} D\pi^{\alpha\beta} + \pi^{\mu\nu} = 2\eta\sigma^{\mu\nu} + \dots. \quad (2.6)$$

Above, the dots indicate possible second-order terms [16, 22, 23] which will be neglected in the following linear stability analysis. Note that most (but not all) second-order terms are nonlinear and do not contribute in the linearized regime (at least when considering perturbations around an equilibrium state). For an analysis of the influence of some of the second-order terms that do contribute in the linear stability analysis, see Ref. [24].

We consider perturbations of all fluid-dynamical fields around the global-equilibrium state described above,

$$\varepsilon = \varepsilon_0 + \delta\varepsilon, \quad (2.7)$$

$$n = n_0 + \delta n, \quad (2.8)$$

$$u^\mu = u_0^\mu + \delta u^\mu, \quad (2.9)$$

$$\Pi = \delta\Pi, \quad (2.10)$$

$$n^\mu = \delta n^\mu, \quad (2.11)$$

$$\pi^{\mu\nu} = \delta\pi^{\mu\nu}, \quad (2.12)$$

where  $n_0 \equiv n(\beta_0, \alpha_0)$  and  $\varepsilon_0 \equiv \varepsilon(\beta_0, \alpha_0)$  are constants. The linearization procedure consists of substituting Eqs. (2.7) – (2.12) into the exact fluid-dynamical equations (2.1) – (2.6) and only retaining the terms that are linear, i.e., of first order, in the deviations from the equilibrium state.

Since the fluid 4-velocity is normalized, i.e.,  $u_\mu u^\mu = 1$ , it is straightforward to demonstrate that the perturbations of the fluid velocity satisfy

$$\delta u_\mu u_0^\mu = \mathcal{O}_2 \approx 0, \quad (2.13)$$

where  $\mathcal{O}_2$  denotes terms that are of second order or higher in perturbations of the fluid-dynamical fields. That is, up to first order in perturbations, the fluctuations of the fluid



4-velocity are orthogonal to the background 4-velocity. Similarly, it is possible to obtain orthogonality relations satisfied by the dissipative currents,  $\delta n^\mu$  and  $\delta \pi^{\mu\nu}$ . As shown in the previous chapter, the dissipative currents are constructed to be orthogonal to the 4-velocity field,  $u_\mu n^\mu = u_\mu \pi^{\mu\nu} = 0$ . Since  $n^\mu$  and  $\pi^{\mu\nu}$  are at least of first order (in this power-counting scheme), these relations lead to

$$u_\mu^0 \delta \pi^{\mu\nu} = \mathcal{O}_2^\nu \approx 0, \quad (2.14)$$

$$u_\mu^0 \delta n^\mu = \mathcal{O}_2 \approx 0. \quad (2.15)$$

Thus, the perturbations of the dissipative currents are, to first order, also orthogonal to the background velocity field. Due to these orthogonality relations, it is convenient to introduce a projection operator onto the 3-space orthogonal to the **background** velocity,

$$\Delta_0^{\mu\nu} \equiv g^{\mu\nu} - u_0^\mu u_0^\nu, \quad (2.16)$$

and, similarly, a rank-4 symmetric, traceless projection operator

$$\Delta_0^{\mu\nu\alpha\beta} \equiv \frac{1}{2} \left( \Delta_0^{\mu\alpha} \Delta_0^{\nu\beta} + \Delta_0^{\mu\beta} \Delta_0^{\nu\alpha} \right) - \frac{1}{3} \Delta_0^{\mu\nu} \Delta_0^{\alpha\beta}. \quad (2.17)$$

Finally, we define a comoving derivative relative to the background velocity field,  $D_0 \equiv u_0^\mu \partial_\mu$ .

Using this notation, the equations of motion for the energy density, local velocity field, and net-charge density, linearized around the global-equilibrium state, become

$$D_0 \delta \varepsilon + (\varepsilon_0 + P_0) \partial_\mu \delta u^\mu = \mathcal{O}_2 \approx 0, \quad (2.18)$$

$$(\varepsilon_0 + P_0) D_0 \delta u^\mu - \Delta_0^{\mu\nu} \partial_\nu (\delta P + \delta \Pi) + \partial_\nu \delta \pi^{\mu\nu} = \mathcal{O}_2^\mu \approx 0, \quad (2.19)$$

$$D_0 \delta n + n_0 \partial_\mu \delta u^\mu + \partial_\mu \delta n^\mu = \mathcal{O}_2 \approx 0, \quad (2.20)$$

while the linearized equations for the dissipative currents are

$$\tau_\Pi D_0 \delta \Pi + \delta \Pi + \zeta \partial_\mu \delta u^\mu = \mathcal{O}_2 \approx 0, \quad (2.21)$$

$$\tau_n D_0 \delta n^\mu + \delta n^\mu - \kappa \Delta_0^{\mu\nu} \partial_\nu \delta \alpha = \mathcal{O}_2^\mu \approx 0, \quad (2.22)$$

$$\tau_\pi D_0 \delta \pi^{\mu\nu} + \delta \pi^{\mu\nu} - 2\eta \Delta_0^{\mu\nu\alpha\beta} \partial_\alpha \delta u_\beta = \mathcal{O}_2^{\mu\nu} \approx 0, \quad (2.23)$$

where  $\zeta$ ,  $\kappa$ ,  $\eta$ ,  $\tau_\Pi$ ,  $\tau_n$ , and  $\tau_\pi$  are the transport coefficients as functions of the *background* temperature and thermal potential. We note that, in the linear regime, pressure perturbations can be expressed as

$$\delta P = \left. \frac{\partial P_0}{\partial \alpha_0} \right|_{\beta_0} \delta \alpha + \left. \frac{\partial P_0}{\partial \beta_0} \right|_{\alpha_0} \delta \beta + \mathcal{O}_2 = \frac{n_0}{\beta_0} \delta \alpha - \frac{h_0 n_0}{\beta_0} \delta \beta + \mathcal{O}_2, \quad (2.24)$$

where we have used Eq. (1.103). Similar relations apply to energy-density and net-charge density fluctuations.

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**Exercise 2.1:** Prove Eqs. (2.18) – (2.23).

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## 2.2 Linearized fluid-dynamical equations in Fourier space

In order to investigate the (propagating and exponentially in- or decreasing) modes of the fluid-dynamical equations, it is convenient to Fourier transform the space-time dependence of the linearized equations. We define our Fourier transformation using the following convention,

$$\tilde{A}(K) = \int d^4x \exp(-ix_\mu k^\mu) A(X) , \quad (2.25)$$

$$A(X) = \int \frac{d^4k}{(2\pi)^4} \exp(ix_\mu k^\mu) \tilde{A}(K) . \quad (2.26)$$

Here,  $k^\mu \equiv K = (\omega, \mathbf{k})^T$  is the wavenumber 4-vector, with  $\omega$  being the frequency and  $\mathbf{k}$  the wavenumber 3-vector of the fluctuation, and  $x^\mu \equiv X = (t, \mathbf{x})^T$  is the usual space-time coordinate 4-vector.

Let us define the covariant variables

$$\Omega \equiv u_0^\mu k_\mu , \quad \kappa^\mu \equiv \Delta^{\mu\nu} k_\nu , \quad (2.27)$$

where  $\Omega$  is the **frequency of the perturbations in the local rest frame of the background fluid** while  $\kappa^\mu$  is the corresponding **wavenumber 4-vector**. The linearized conservation laws (2.18) – (2.20) can then be written in Fourier space in the simple form

$$\Omega \delta \tilde{\varepsilon} + (\varepsilon_0 + P_0) \kappa_\mu \delta \tilde{u}^\mu = 0 , \quad (2.28)$$

$$(\varepsilon_0 + P_0) \Omega \delta \tilde{u}^\mu - \kappa^\mu \left( \delta \tilde{P} + \delta \tilde{\Pi} \right) + \kappa_\nu \delta \tilde{\pi}^{\mu\nu} = 0 , \quad (2.29)$$

$$\Omega \delta \tilde{n} + n_0 \kappa_\mu \delta \tilde{u}^\mu + \kappa_\mu \delta \tilde{n}^\mu = 0 , \quad (2.30)$$

and the linearized equations of motion (2.21) – (2.23) for the dissipative currents become

$$(1 + i\tau_\Pi \Omega) \delta \tilde{\Pi} = -i\zeta \kappa_\mu \delta \tilde{u}^\mu , \quad (2.31)$$

$$(1 + i\tau_n \Omega) \delta \tilde{n}^\mu = i\chi \kappa^\mu \delta \tilde{\alpha} , \quad (2.32)$$

$$(1 + i\tau_\pi \Omega) \delta \tilde{\pi}^{\mu\nu} = i\eta \left( \kappa^\mu \delta \tilde{u}^\nu + \kappa^\nu \delta \tilde{u}^\mu - \frac{2}{3} \Delta_0^{\mu\nu} \kappa_\lambda \delta \tilde{u}^\lambda \right) . \quad (2.33)$$

We further define the scalar  $\kappa$  as the modulus of  $\kappa^\mu$ ,  $\kappa_\mu \kappa^\mu \equiv -\kappa^2$ . Note that the usual dispersion relation of perturbations in a fluid,  $\omega(\mathbf{k})$  (with  $\omega \equiv k^0$ ), is not a Lorentz scalar and will change depending on the magnitude of the background velocity  $u_0^\mu$ .

### 2.2.1 Tensor decomposition in Fourier space

It is also convenient to decompose the perturbations into components **parallel** and **orthogonal** to  $\kappa^\mu$ . For this purpose we introduce another **projection operator**, now onto the 3-space orthogonal to  $\hat{\kappa}^\mu$ ,

$$\Delta_\kappa^{\mu\nu} = g^{\mu\nu} + \hat{\kappa}^\mu \hat{\kappa}^\nu , \quad (2.34)$$

where  $\hat{\kappa}^\mu \equiv \kappa^\mu / \kappa$  is a space-like 4-vector normalized to  $\hat{\kappa}_\mu \hat{\kappa}^\mu = -1$ . Within this scheme, an arbitrary 4-vector  $A^\mu$  is decomposed as

$$A^\mu = A_{\parallel} \hat{\kappa}^\mu + A_{\perp}^\mu, \quad (2.35)$$

with components  $A_{\parallel} \equiv -\hat{\kappa}_\mu A^\mu$  and  $A_{\perp}^\mu \equiv \Delta_{\kappa}^{\mu\lambda} A_\lambda$ . We shall refer to  $A_{\parallel}$  as the **longitudinal** component of the corresponding 4-vector and  $A_{\perp}^\mu$  as its **transverse** component. Similarly, an arbitrary traceless, symmetric second-rank tensor,  $A^{\mu\nu}$ , is decomposed as

$$A^{\mu\nu} = A_{\parallel} \hat{\kappa}^\mu \hat{\kappa}^\nu + \frac{A_{\parallel}}{3} \Delta_{\kappa}^{\mu\nu} + A_{\perp}^\mu \hat{\kappa}^\nu + A_{\perp}^\nu \hat{\kappa}^\mu + A_{\perp}^{\mu\nu}. \quad (2.36)$$

with  $A_{\parallel} \equiv \hat{\kappa}^\mu \hat{\kappa}^\nu A_{\mu\nu}$ ,  $A_{\perp}^\lambda \equiv -\hat{\kappa}^\nu \Delta_{\kappa}^{\mu\lambda} A_{\mu\nu}$ , and  $A_{\perp}^{\mu\nu} \equiv \Delta_{\kappa}^{\mu\nu\alpha\beta} A_{\alpha\beta}$ , where we defined the rank-4 symmetric, traceless projection operator  $\Delta_{\kappa}^{\mu\nu\alpha\beta} \equiv \frac{1}{2}(\Delta_{\kappa}^{\mu\alpha} \Delta_{\kappa}^{\nu\beta} + \Delta_{\kappa}^{\mu\beta} \Delta_{\kappa}^{\nu\alpha}) - \frac{1}{3} \Delta_{\kappa}^{\mu\nu} \Delta_{\kappa}^{\alpha\beta}$ . In this case,  $A_{\parallel}$  is the **longitudinal** component of the tensor,  $A_{\perp}^\mu$  corresponds to its **partially transverse** component, and  $A_{\perp}^{\mu\nu}$  is its **fully transverse** (traceless) component.

## 2.2.2 Longitudinal and transverse components

We now project Eqs. (2.28) – (2.33) onto their components longitudinal and transverse to  $\hat{\kappa}^\mu$ . This is useful because the longitudinal and transverse projections of the equations **decouple** from each other, and can be solved **independently** to obtain the dispersion relations satisfied by the perturbations. We first consider the equations for the longitudinal components of the fluctuations.

First, we note that Eqs. (2.28), (2.30), and (2.31) are already written in terms of longitudinal perturbations since they describe perturbations of scalar quantities, i.e., energy density, net-charge density, and bulk-viscous pressure. The longitudinal component of Eqs. (2.29) and (2.32) are obtained by projecting each equation onto  $\hat{\kappa}_\mu$ , while the longitudinal component of Eq. (2.33) is obtained by projecting it onto  $\hat{\kappa}_\mu \hat{\kappa}_\nu$ . The result is a set of coupled equations for the perturbations  $\delta\tilde{\varepsilon}$ ,  $\delta\tilde{P}$ ,  $\delta\tilde{n}$ ,  $\delta\tilde{\alpha}$ ,  $\delta\tilde{u}_{\parallel}$ ,  $\delta\tilde{\Pi}$ ,  $\delta\tilde{n}_{\parallel}$ , and  $\delta\tilde{\pi}_{\parallel}$ ,

$$\Omega \delta\tilde{\varepsilon} - \kappa (\varepsilon_0 + P_0) \delta\tilde{u}_{\parallel} = 0, \quad (2.37)$$

$$(\varepsilon_0 + P_0) \Omega \delta\tilde{u}_{\parallel} - \kappa (\delta\tilde{P} + \delta\tilde{\Pi} + \delta\tilde{\pi}_{\parallel}) = 0, \quad (2.38)$$

$$\Omega \delta\tilde{n} - \kappa (n_0 \delta\tilde{u}_{\parallel} + \delta\tilde{n}_{\parallel}) = 0, \quad (2.39)$$

$$(1 + i\tau_{\Pi}\Omega) \delta\tilde{\Pi} - i\zeta\kappa\delta\tilde{u}_{\parallel} = 0, \quad (2.40)$$

$$(1 + i\tau_n\Omega) \delta\tilde{n}_{\parallel} - i\kappa\delta\tilde{\alpha} = 0, \quad (2.41)$$

$$(1 + i\tau_{\pi}\Omega) \delta\tilde{\pi}_{\parallel} - \frac{4}{3}i\eta\kappa\delta\tilde{u}_{\parallel} = 0. \quad (2.42)$$

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**Exercise 2.2:** Prove Eqs. (2.37) – (2.42).

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The fluctuations of energy density, net-charge density, and pressure can be converted into fluctuations of inverse temperature and thermal potential,  $\delta\tilde{\beta}$  and  $\delta\tilde{\alpha}$ , respectively. Solving these equations leads to **six different modes** of the theory, two of them related

to the **propagation of sound waves**, one related to the **diffusion of net-charge density fluctuations**, and the remaining three being new non-propagating modes due to **fluctuations of the dissipative currents** that are not allowed in Navier-Stokes theory. The latter modes are **non-hydrodynamic**, i.e., they are modes that do not vanish when the wavenumber is taken to zero, and describe the **relaxation of the dissipative currents back to global equilibrium**. Such modes were thought to never exist in fluid-dynamical theories, but we shall see that they are **required in the relativistic regime in order to restore causality** – and that they even appear in Navier-Stokes theory in the relativistic regime.

The equation of motion for the transverse component of the velocity field is obtained by projecting Eq. (2.29) with  $\Delta_{\kappa\mu}^\lambda$  while the equation of motion for the partially transverse component of the shear-stress tensor is obtained by projecting Eq. (2.33) with  $\hat{\kappa}_\mu \Delta_{\kappa\nu}^\lambda$ . The resulting equations are coupled,

$$(\varepsilon_0 + P_0) \Omega \delta \tilde{u}_\perp^\lambda - \kappa \delta \tilde{\pi}_\perp^\lambda = 0 , \quad (2.43)$$

$$(1 + i\tau_\pi \Omega) \delta \tilde{\pi}_\perp^\lambda - i\eta \kappa \delta \tilde{u}_\perp^\lambda = 0 . \quad (2.44)$$

**Exercise 2.3:** Prove Eqs. (2.43) and (2.44).

Such transverse modes are usually referred to as **shear modes**, since they do not display any contributions from bulk-viscous pressure or diffusion 4-current. They describe the **diffusion of the velocity field** and, as long as the relaxation times remain finite, they also contain a non-hydrodynamic mode. Since  $\delta \tilde{u}_\perp^\mu$  and  $\delta \tilde{\pi}_\perp^\mu$  carry a total of four independent degrees of freedom, each mode obtained from the above equations will have a twofold degeneracy.

Finally, we have the two equations of motion for the transverse fluctuation of the diffusion 4-current and the fully transverse fluctuation of the shear-stress tensor, also containing two modes each,

$$(1 + i\tau_n \Omega) \delta \tilde{n}_\perp^\lambda = 0 , \quad (2.45)$$

$$(1 + i\tau_\pi \Omega) \delta \tilde{\pi}_\perp^{\alpha\beta} = \frac{2}{9} i \eta \kappa \delta \tilde{u}_\parallel \left( \Delta_\kappa^{\alpha\beta} - 3 u_0^\alpha u_0^\beta \right) . \quad (2.46)$$

The first equation is obtained by projecting Eq. (2.32) with  $\Delta_{\kappa\mu}^\lambda$  while the second equation is obtained by projecting Eq. (2.33) with  $\Delta_{\kappa\mu\nu}^{\alpha\beta}$ .

**Exercise 2.4:** Prove Eqs. (2.45) and (2.46).

Equation (2.45) gives rise to a twofold degenerate mode,

$$\Omega = \frac{i}{\tau_n} , \quad (2.47)$$

which, in the rest frame of the global-equilibrium background,  $u_0^\mu = (1, 0, 0, 0)^T$ , has no wavenumber dependence, i.e., it is non-propagating. Such a mode is obviously **non-hydrodynamic**, i.e., it has a frequency that does not vanish when the wavenumber is taken to zero in the local rest frame. It is purely imaginary, with positive imaginary part, and thus related to the exponential damping of the dissipative currents towards the equilibrium state. Such modes are rather simple and will not be discussed further.

Equation (2.46) also gives rise to a twofold degenerate mode. With the help of Eq. (2.42) we can write this equation as

$$\delta\tilde{\pi}_\perp^{\alpha\beta} = \frac{1}{6} \delta\tilde{\pi}_\parallel \left( \Delta_\kappa^{\alpha\beta} - 3u_0^\alpha u_0^\beta \right), \quad (2.48)$$

which shows that the fully transverse fluctuations of  $\pi^{\alpha\beta}$  simply follow the time dependence of the longitudinal one. In the next sections, we discuss the solutions of the longitudinal modes and the remaining transverse modes for ideal fluids, viscous fluids in the Navier-Stokes limit, and for the complete transient fluid-dynamical equations.

## 2.3 Ideal fluid dynamics

The ideal-fluid limit is obtained by setting the viscosity and relaxation-time coefficients to zero. In this case, one obtains the following set of equations for the longitudinal perturbations

$$\Omega \delta\tilde{\varepsilon} - (\varepsilon_0 + P_0) \kappa \delta\tilde{u}_\parallel = 0, \quad (2.49)$$

$$(\varepsilon_0 + P_0) \Omega \delta\tilde{u}_\parallel - \kappa \delta\tilde{P} = 0, \quad (2.50)$$

$$\Omega \delta\tilde{n} - n_0 \kappa \delta\tilde{u}_\parallel = 0, \quad (2.51)$$

and one equation for the transverse velocity perturbation

$$(\varepsilon_0 + P_0) \Omega \delta\tilde{u}_\perp^\mu = 0. \quad (2.52)$$

The set of equations for the longitudinal perturbations can be cast into the following matrix form,

$$\begin{pmatrix} \Omega \left. \frac{\partial \varepsilon_0}{\partial \beta_0} \right|_{\alpha_0} & \Omega \left. \frac{\partial \varepsilon_0}{\partial \alpha_0} \right|_{\beta_0} & -(\varepsilon_0 + P_0) \kappa \\ \kappa \frac{h_0 n_0}{\beta_0} & -\kappa \frac{n_0}{\beta_0} & (\varepsilon_0 + P_0) \Omega \\ \Omega \left. \frac{\partial n_0}{\partial \beta_0} \right|_{\alpha_0} & \Omega \left. \frac{\partial n_0}{\partial \alpha_0} \right|_{\beta_0} & -n_0 \kappa \end{pmatrix} \begin{pmatrix} \delta\tilde{\beta} \\ \delta\tilde{\alpha} \\ \delta\tilde{u}_\parallel \end{pmatrix} = 0, \quad (2.53)$$

where the energy-density, net-charge density, and pressure perturbations were decomposed in terms of inverse temperature and thermal-potential perturbations. Note that, for the pressure perturbation, we have used Eq. (2.24).

The solution for the transverse mode is simply  $\Omega = 0$ . The remaining modes from the longitudinal fluctuations are obtained by finding the roots of the determinant of the matrix in Eq. (2.53), leading to the following equation

$$\Omega (\Omega^2 - c_s^2 \kappa^2) = 0, \quad (2.54)$$

where we identify the square of the **sound velocity** as

$$c_s^2 = \frac{\partial P_0}{\partial \varepsilon_0} \Big|_{s_0/n_0} = \frac{n_0}{\beta_0} \frac{h_0^{-1} \frac{\partial \varepsilon_0}{\partial \beta_0} \Big|_{\alpha_0} - \frac{\partial n_0}{\partial \beta_0} \Big|_{\alpha_0} + \frac{\partial \varepsilon_0}{\partial \alpha_0} \Big|_{\beta_0} - h_0 \frac{\partial n_0}{\partial \alpha_0} \Big|_{\beta_0}}{\frac{\partial \varepsilon_0}{\partial \beta_0} \Big|_{\alpha_0} \frac{\partial n_0}{\partial \alpha_0} \Big|_{\beta_0} - \frac{\partial \varepsilon_0}{\partial \alpha_0} \Big|_{\beta_0} \frac{\partial n_0}{\partial \beta_0} \Big|_{\alpha_0}} . \quad (2.55)$$

**Exercise 2.5:**

- (i) Prove Eq. (2.54) by explicitly computing the determinant of the coefficient matrix in Eq. (2.53).
- (ii) Prove Eq. (2.55) by applying the chain rule for Jacobi determinants, i.e.,

$$\frac{\partial(x, y)}{\partial(u, v)} = \frac{\partial(x, y)}{\partial(a, b)} \frac{\partial(a, b)}{\partial(u, v)} . \quad (2.56)$$

In the **rest frame** of the global-equilibrium state, i.e.,  $u_0^\mu = (1, 0, 0, 0)^T$ , the scalars  $\Omega$  and  $\kappa$  reduce to the usual frequency and wavenumber of the perturbations, respectively,  $\Omega = \omega$  and  $\kappa = k$ . In this case, one recovers the well-known expressions for the modes of an ideal fluid,

$$\omega(k) = 0 , \quad \omega(k) = \pm c_s k , \quad (2.57)$$

i.e., a **static mode** and two **sound modes**, travelling with velocity  $c_s$  forward and backward with respect to the fluid at rest. The static mode appears because we included net-charge fluctuations in the analysis. If one considers a system with zero net charge, the static mode disappears, with the sound modes remaining in the same form as obtained above.

For a **nonzero background velocity** (without loss of generality, we can always set the velocity to point into the  $x$ -direction),  $u_0^\mu = (\gamma, \gamma V, 0, 0)^T$ , we have  $\Omega = \gamma\omega - \gamma V k^x$  and  $\kappa^2 = \Omega^2 - k_\mu k^\mu = \gamma^2 (\omega V - k^x)^2 + k_T^2$ . Naturally, a nonzero background velocity introduces an **anisotropy** in the system, with the modes no longer depending on the modulus of the wavenumber, but displaying a separate dependence on  $k^x$  and  $k_T \equiv \sqrt{k_y^2 + k_z^2}$ . For the sake of simplicity, we consider **waves traveling only in the  $x$ -direction**, i.e., with  $k_T = 0$ , in which case the frequency will depend solely on  $k \equiv k^x$ . Then the dispersion relation is obtained by solving the equation

$$(\omega - V k) \left[ (1 - c_s^2 V^2) \omega^2 - 2(1 - c_s^2) V k \omega + (V^2 - c_s^2) k^2 \right] = 0 . \quad (2.58)$$

This equation has three solutions,

$$\omega(k) = V k , \quad (2.59)$$

$$\omega_\pm(k) = \frac{V \pm c_s}{1 \pm V c_s} k . \quad (2.60)$$

We see that the frequencies of oscillations are both linear in  $k$ . As expected, the mode that previously vanished now starts to move with the background velocity. Meanwhile, each sound mode has now a different propagation speed, expressed by the **relativistic velocity-addition rule**

$$v_s^\pm = \frac{V \pm c_s}{1 \pm V c_s} .$$

One mode is moving **in** the direction of the unperturbed fluid and the other in the **opposite** direction. Naturally, the waves propagating against the flow of the background will propagate slower than those moving in the same direction as the background fluid.

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**Exercise 2.6:** Check that Eq. (2.60) is indeed a solution of Eq. (2.58).

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05/11/2022

## 2.4 Relativistic Navier-Stokes theory

The modes related to relativistic Navier-Stokes theory are obtained from the expressions derived in the previous section by taking the limit of vanishing relaxation times,  $\tau_\pi = \tau_n = \tau_\Pi = 0$ . In this limit, Eqs. (2.40), (2.41), (2.42), and (2.44) dictate that the fluctuations of the dissipative currents are expressed in terms of fluctuations of velocity and thermal potential in the following way,

$$\delta\tilde{\Pi} = i\zeta\kappa\delta\tilde{u}_\parallel , \quad (2.61)$$

$$\delta\tilde{n}_\parallel = i\kappa\delta\tilde{\alpha} , \quad (2.62)$$

$$\delta\tilde{\pi}_\parallel = \frac{4}{3}i\eta\kappa\delta\tilde{u}_\parallel , \quad (2.63)$$

$$\delta\tilde{\pi}_\perp^\lambda = i\eta\kappa\delta\tilde{u}_\perp^\lambda . \quad (2.64)$$

The equations of motion (2.37) – (2.39) for the longitudinal perturbations then reduce to

$$\Omega\delta\tilde{\varepsilon} - \kappa(\varepsilon_0 + P_0)\delta\tilde{u}_\parallel = 0 , \quad (2.65)$$

$$(\varepsilon_0 + P_0)\Omega\delta\tilde{u}_\parallel - \kappa\delta\tilde{P} - i\kappa^2\left(\zeta + \frac{4}{3}\eta\right)\delta\tilde{u}_\parallel = 0 , \quad (2.66)$$

$$\Omega\delta\tilde{n} - \kappa n_0\delta\tilde{u}_\parallel - i\kappa^2\delta\tilde{\alpha} = 0 , \quad (2.67)$$

while the transverse modes (2.43) satisfy the simple relation

$$\left(\Omega - i\kappa^2\frac{\eta}{\varepsilon_0 + P_0}\right)\delta\tilde{u}_\perp^\mu = 0 . \quad (2.68)$$

There are no transverse perturbations of the diffusion current, while on account of Eq. (2.48) the fully transverse perturbations of the shear-stress tensor follow that of the longitudinal one, Eq. (2.63).

### 2.4.1 Transverse modes

Let us first discuss the transverse (i.e., shear) modes since their dispersion relations are more simple and easier to obtain. From Eq. (2.68), we can immediately infer that  $\Omega$  and  $\kappa$  obey a **diffusion-type dispersion relation**

$$\Omega = i\tau_\eta \kappa^2 . \quad (2.69)$$

where we defined

$$\tau_\eta \equiv \frac{\eta}{\varepsilon_0 + P_0} . \quad (2.70)$$

We note that  $\tau_\eta$  is a time scale that naturally appears in the shear modes of Navier-Stokes theory and encompasses all effects of dissipation on the transverse perturbations. We shall soon demonstrate that this time scale is also related to the exponential growth of unstable non-hydrodynamic modes that will appear when perturbing the theory in a moving background. These unstable modes appear only due to the parabolic nature of the relativistic Navier-Stokes theory, which ends up generating new modes in boosted frames.

When  $u_0^\mu = (1, 0, 0, 0)^T$ , one simply obtains a non-propagating, exponentially damped mode with twofold degeneracy,

$$\omega(k) = i\tau_\eta k^2 . \quad (2.71)$$

This relation is the same as the one obtained by solving the diffusion equation, with  $\tau_\eta$  playing the role of the diffusion coefficient. That is, the shear modes simply describes a diffusion process of the velocity field.

Now we consider propagation of shear perturbations in a moving background. As done for the case of an ideal fluid, we choose our coordinate system in such a way that the velocity of the background is in the  $x$ -direction, i.e.,  $u_0^\mu = (\gamma, \gamma V, 0, 0)^T$ . We also only consider the case of perturbations that travel in the  $x$ -direction, i.e.,  $k^\mu = (\omega, k, 0, 0)^T$ . The dispersion relation (2.69) then becomes

$$\begin{aligned} \gamma(\omega - Vk) &= i\tau_\eta \gamma^2 (\omega V - k)^2 \\ \iff 0 &= \omega^2 V^2 + \left( \frac{i}{\gamma\tau_\eta} - 2Vk \right) \omega - \frac{i}{\gamma\tau_\eta} Vk + k^2 . \end{aligned} \quad (2.72)$$

A **nonzero background velocity** has the effect of **mixing the contributions of frequency,  $\omega$ , and wavenumber,  $k$** , which are contained in the Lorentz scalars  $\Omega$  and  $\kappa$ . Therefore, for a moving background the quadratic term in  $\kappa$  in the dispersion relation will not only carry contributions that are quadratic in  $k$ , as in the  $V = 0$  case, but will also carry contributions that are **quadratic** in  $\omega$ . This leads to the appearance of the quadratic term  $\omega^2 V^2$  in the dispersion relation, producing an **additional transverse mode** in the theory. The two solutions are

$$\omega(k) = \frac{1 + 2i\gamma V\tau_\eta k \pm \sqrt{1 + 4iV\tau_\eta k/\gamma}}{2i\gamma V^2\tau_\eta} . \quad (2.73)$$

The analytic solution in the limit of small wavenumbers,  $k \rightarrow 0$ , is

$$\omega(k \rightarrow 0) = \begin{cases} -\frac{i}{\gamma V^2 \tau_\eta} , \\ Vk + i\tau_\eta k^2 / \gamma^3 . \end{cases} \quad (2.74)$$



The second mode corresponds to the one of Eq. (2.71), since it tends to  $ik^2\tau_\eta$  when the background velocity goes to zero. On the other hand, the first mode is intrinsically new and is non-hydrodynamic, non-propagating, and, because of a negative imaginary part, **unstable**. This mode describes perturbations that **grow exponentially** on time scales of the order of  $\gamma V^2\tau_\eta$ . This additional mode does **not** appear when the velocity of the unperturbed fluid is **zero** and, for this reason, this problematic feature of Navier-Stokes theory remains largely unperceived in the literature. Nevertheless, the emergence of unstable non-hydrodynamic modes in relativistic Navier-Stokes theory is a fundamental issue that must be fixed, in order to obtain a relativistic theory of fluid dynamics. We remark that this is a problem of the relativistic version of Navier-Stokes theory – the nonrelativistic version of the theory is stable, even when perturbed around a **moving** global-equilibrium state.

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**Exercise 2.7:** Prove Eqs. (2.73) and (2.74).

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## 2.4.2 Longitudinal modes

Next, we consider the longitudinal modes. For the sake of simplicity, we restrict our analysis to the limit of vanishing net-charge fluctuations. (This case is treated in Ref. [24] for a vanishing net background charge). In this case, Eqs. (2.65) and (2.66) simplify to

$$\begin{pmatrix} \Omega & -\kappa \\ -c_s^2\kappa & \Omega - i\tau_{\text{eff}}\kappa^2 \end{pmatrix} \begin{pmatrix} \frac{\delta\tilde{\varepsilon}}{\varepsilon_0 + P_0} \\ \delta\tilde{u}_\parallel \end{pmatrix} = 0, \quad (2.75)$$

where we used the definition of sound velocity at zero chemical potential,  $c_s^2 = dP_0/d\varepsilon_0$ , and defined the effective time scale

$$\tau_{\text{eff}} \equiv \frac{\zeta + \frac{4}{3}\eta}{\varepsilon_0 + P_0}, \quad (2.76)$$

which plays a very similar role as the variable  $\tau_\eta$  that appeared in the previous subsection when discussing the shear modes. As before, the dispersion relations satisfied by the perturbations are found by setting the determinant of the matrix in Eq. (2.75) to zero. The equation satisfied by  $\Omega$  and  $\kappa$  then is

$$\Omega^2 - i\tau_{\text{eff}}\Omega\kappa^2 - c_s^2\kappa^2 = 0. \quad (2.77)$$

In the case where the velocity of the unperturbed system is zero, i.e.,  $u_0^\mu = (1, 0, 0, 0)^T$ , one simply has  $\Omega = \omega$  and  $\kappa = k$  and Eq. (2.77) simplifies to

$$\omega^2 - i\tau_{\text{eff}}\omega k^2 - c_s^2 k^2 = 0 \implies \omega(k) = i\frac{\tau_{\text{eff}}}{2}k^2 \pm k\sqrt{c_s^2 - \frac{\tau_{\text{eff}}^2 k^2}{4}}. \quad (2.78)$$

These are the sound modes of the theory and they reduce to the solution found for ideal fluids when  $\tau_{\text{eff}} = 0$ . Defining the critical wavenumber

$$k_c^{\text{NS}} = 2c_s/\tau_{\text{eff}}, \quad (2.79)$$

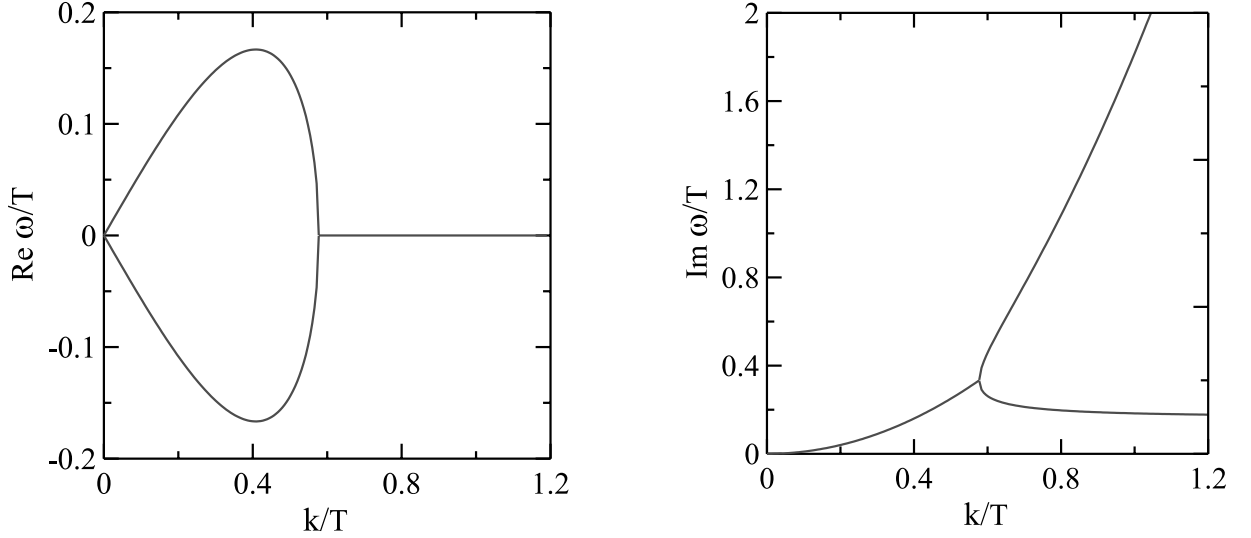


Figure 2.1: The real parts (left panel) and the imaginary parts (right panel) of the dispersion relations for the longitudinal modes of Navier-Stokes theory for a static background. We set  $\tau_{\text{eff}} = 2\beta_0$ .

we see that for  $k < k_c^{\text{NS}}$  the sound modes have a part propagating with the reduced speed of sound  $c_{s,\text{eff}} \equiv \sqrt{c_s^2 - \tau_{\text{eff}}^2 k^2/4} \leq c_s$ . They also have a non-propagating part which describes how such modes are damped by viscosity. On the other hand, for wavenumbers larger than  $k \geq k_c^{\text{NS}}$ , the frequencies become purely imaginary and there are no propagating modes. In Fig. 2.1 we show the real (left panel) and imaginary (right panel) parts of  $\omega(k)$  in units of temperature, for  $\tau_{\text{eff}} = 2\beta_0$ .

The solution at small wavenumbers,  $k \rightarrow 0$ , can be written in the simple form

$$\omega(k) = \pm c_s k + i \frac{\tau_{\text{eff}}}{2} k^2 + \mathcal{O}(k^3), \quad (2.80)$$

from which we conclude that, to leading order, the sound modes propagate with the usual speed of sound  $c_s$ , damped by the effective specific viscosity  $\tau_{\text{eff}}/2$ . On the other hand, at large wavenumbers,  $k \rightarrow \infty$ ,

$$\omega(k) = i \frac{\tau_{\text{eff}}}{2} k^2 (1 \pm 1) + \mathcal{O}(1). \quad (2.81)$$

That is, for large values of wavenumbers, the mode is either zero or appears to be purely diffusive,  $\omega(k) \sim i \tau_{\text{eff}} k^2$ .

For the same moving background as considered for the shear modes,  $u_0^\mu = \gamma(1, V, 0, 0)^T$  and  $k^\mu = (\omega, k, 0, 0)^T$ , the dispersion relation (2.77) assumes the form

$$(\omega - kV)^2 - i\tau_{\text{eff}}\gamma(\omega - kV)(\omega V - k)^2 - c_s^2(\omega V - k)^2 = 0. \quad (2.82)$$

In this case, we observe that the term  $\Omega\kappa^2$  in Eq. (2.77) will make the equation **cubic** in  $\omega$  and, as happened with the shear modes, a new solution will emerge. It is easy to see that such a new solution will be an **unstable non-hydrodynamic mode**. For this

purpose, one can just look for the solutions of the equation at  $k = 0$ ,

$$\omega(k=0) = \begin{cases} -i \frac{1 - c_s^2 V^2}{\tau_{\text{eff}} \gamma V^2} , \\ \pm 0 . \end{cases} \quad (2.83)$$

The last two solutions are simply zero and can be identified with the usual sound modes that already appeared when the unperturbed global-equilibrium state was at rest. The additional solution does not vanish at  $k = 0$ , i.e., it is a non-hydrodynamic mode, and has a negative imaginary part, i.e., it is an unstable exponentially growing mode. Once more, we see that **relativistic Navier-Stokes theory is unstable** when perturbed around global equilibrium: a fundamental problem that simply cannot be ignored. In the next subsection, we briefly discuss the origins behind this issue.

### 2.4.3 Causality and stability of Navier-Stokes theory

In the preceding subsections it was shown that relativistic Navier-Stokes theory is unstable, since perturbations on a moving fluid in global equilibrium grow exponentially, within microscopic time scales of the order of  $\tau_\eta \sim \tau_{\text{eff}} \sim \eta/(\varepsilon_0 + P_0)$ . This analysis was performed fixing the direction of the background velocity to be in the  $x$ -direction and only considering perturbations traveling in the same direction, i.e.,

$$\begin{aligned} u_0^\mu &= \gamma(1, V, 0, 0)^T , \\ k^\mu &= (\omega, k, 0, 0)^T . \end{aligned} \quad (2.84)$$

In this scenario, the covariant frequency,  $\Omega$ , and wavenumber,  $\kappa$ , of the oscillations satisfy

$$\Omega = \gamma(\omega - V k) , \quad (2.85)$$

$$\kappa^2 = \gamma^2(\omega V - k)^2 , \quad (2.86)$$

which is equivalent to a 1+1-dimensional Lorentz boost with velocity  $V$  of a 4-vector made of  $\omega$  and  $k$ . Therefore, for all practical purposes, the dispersion relation for modes obtained for perturbations of a moving fluid can be obtained by applying a Lorentz transformation to the same dispersion relation obtained for perturbations on a fluid at rest. This can help us understand the connection between the acausal nature of the theory and its perturbative instability when considering moving background fluids. We shall make this argument also noting a connection between the modes of Navier-Stokes theory, at asymptotically large values of wavenumber, and the dispersion relation of the diffusion equation.

We already demonstrated in the previous subsections that, for perturbations of a static fluid, both the transverse and longitudinal modes of Navier-Stokes theory behave in the same way when  $k \rightarrow \infty$ ,  $\omega(k) \sim i\tau_{\text{eff}}k^2/2 \sim i\tau_\eta k^2$ , being purely non-propagating modes that are quadratic in  $k$ . This behavior is identical to the one found in the diffusion equation (1.113), in which case there is a single non-propagating mode with dispersion relation  $\omega(k) = iDk^2$ , with  $D$  being the diffusion coefficient. It is well known that the diffusion equation is acausal and, consequently, we conjecture that a  $k^2$ -dependence of any non-propagating mode can also be considered a sign of acausality.

Now we show that the parabolic nature of diffusion-like dispersion relations (given by their quadratic dependence on  $k$ ) will lead to additional modes once a Lorentz transformation is performed. Let us consider the Lorentz boost of a 4-vector with time-component  $\omega$  and spatial component  $k$  in boost direction,

$$\begin{pmatrix} \omega \\ k \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma V \\ -\gamma V & \gamma \end{pmatrix} \begin{pmatrix} \omega' \\ k' \end{pmatrix} = \begin{pmatrix} \gamma\omega' - \gamma V k' \\ -\gamma V \omega' + \gamma k' \end{pmatrix}. \quad (2.87)$$

Then, we substitute the boosted variables into the diffusion dispersion relation, which changes in the following way

$$\omega = iDk^2 \longrightarrow \omega' - V k' = i\gamma D (k' - V \omega')^2. \quad (2.88)$$

It is straightforward to see that, in the limit of vanishing  $k'$ , there is always a **non-hydrodynamic mode** given by

$$\omega'(k' = 0) = -\frac{i}{\gamma D V^2}, \quad (2.89)$$

which has a **negative imaginary part**, i.e., it is **exponentially growing and thus unstable**. Note that all unstable modes obtained so far in Navier-Stokes theory have the structure above, i.e., at  $k = 0$  they always look like boosted modes of the diffusion equation. In this sense, it is not too surprising that a theory with parabolic, diffusion-type modes in a non-moving background will end up featuring unstable modes in a moving background. Naturally, **non-relativistic theories** will **never** display this property since the modes transform following **Galileo's transformation**, which **never mixes frequency and wavenumber**.

It is important to note how the structure of Lorentz boosts ends up connecting the **infinite-wavenumber behavior** of a mode in a **static background** (usually considered to be irrelevant for macroscopic dynamics) with the **vanishing-wavenumber** behavior of the same mode in a **moving background**. This is partially why the **acausality** of a mode, which is an **asymptotic feature**, can affect its **stability**, which is a feature influencing the **small-wavenumber behavior** of the fluctuation. This is certainly an interesting aspect of relativistic theories.

Finally, we shall see in the next section that a transient theory of fluid dynamics does not display any non-propagating modes that match those of the diffusion equation. Also, we shall see that intrinsically unstable modes do not exist – the modes can be rendered stable as long as the transport coefficients satisfy a set of conditions that will be derived below.

## 2.5 Transient theory of fluid dynamics

We now analyze the modes of the complete theory, which includes the effect of the relaxation times. For the sake of simplicity, we shall perform this analysis neglecting contributions from net-charge fluctuations and in the conformal limit (vanishing bulk-viscous

pressure,  $\delta\tilde{\Pi} = 0$ , and vanishing bulk viscosity,  $\zeta = 0$ ). In this case, Eqs. (2.37), (2.38), and (2.42) for the longitudinal fluctuations simplify to

$$\Omega\delta\tilde{\varepsilon} - \kappa(\varepsilon_0 + P_0)\delta\tilde{u}_{\parallel} = 0, \quad (2.90)$$

$$(\varepsilon_0 + P_0)\Omega\delta\tilde{u}_{\parallel} - \kappa(c_s^2\delta\tilde{\varepsilon} + \delta\tilde{\pi}_{\parallel}) = 0, \quad (2.91)$$

$$(1 + i\tau_{\pi}\Omega)\delta\tilde{\pi}_{\parallel} - \frac{4}{3}i\eta\kappa\delta\tilde{u}_{\parallel} = 0. \quad (2.92)$$

The remaining equations for the shear modes do not change under these assumptions and are still given by Eqs. (2.43), (2.44), and (2.46). As already discussed above, the solution of Eq. (2.46) is already determined by Eq. (2.92) and does not need to be addressed further.

### 2.5.1 Transverse modes in the rest frame

The equations of motion (2.43), (2.44) for the transverse (i.e., shear) modes can be cast into the following matrix form

$$\begin{pmatrix} \Omega & -\kappa \\ -i\tau_{\eta}\kappa & 1 + i\tau_{\pi}\Omega \end{pmatrix} \begin{pmatrix} \delta\tilde{u}_{\perp}^{\mu} \\ \frac{\delta\tilde{\pi}_{\perp}^{\mu}}{\varepsilon_0 + P_0} \end{pmatrix} = 0, \quad (2.93)$$

leading to the following the dispersion relation,

$$\Omega(1 + i\tau_{\pi}\Omega) - i\tau_{\eta}\kappa^2 = 0. \quad (2.94)$$

Naturally, this equation will have two solutions in both the moving background and the one at rest, since its highest power in  $\Omega$  is the same as its highest power in  $\kappa$ , i.e.,  $\kappa^2$ .

When the background fluid is at rest, i.e., when we set  $u_0^{\mu} = (1, 0, 0, 0)^T$ , the dispersion relation becomes

$$\omega(1 + i\tau_{\pi}\omega) - i\tau_{\eta}k^2 = 0, \quad (2.95)$$

with two distinct solutions

$$\omega(k) = \frac{i}{2\tau_{\pi}} \pm k\sqrt{\frac{\tau_{\eta}}{\tau_{\pi}}} \sqrt{1 - \left(\frac{\kappa_c^{\text{shear}}}{k}\right)^2}, \quad (2.96)$$

where similarly to Navier-Stokes theory we defined a critical wavenumber

$$\kappa_c^{\text{shear}} \equiv \frac{1}{\sqrt{4\tau_{\eta}\tau_{\pi}}}. \quad (2.97)$$

We see that, due to the nonzero relaxation time  $\tau_{\pi}$ , the transverse shear modes become propagating for  $k > \kappa_c^{\text{shear}}$ , with propagation speed  $\sqrt{\tau_{\eta}/\tau_{\pi}}\sqrt{1 - (\kappa_c^{\text{shear}}/k)^2}$ . They also have a nonvanishing imaginary part, describing damping on a time scale  $\sim 2\tau_{\pi}$ .

At small wavenumbers,  $k \rightarrow 0$ , the shear modes become

$$\omega(k \rightarrow 0) = \begin{cases} i\tau_{\eta}k^2 + \mathcal{O}(k^4), \\ \frac{i}{\tau_{\pi}} + \mathcal{O}(k^2). \end{cases} \quad (2.98)$$

We see that, for transient theories of fluid dynamics, we already have a **non-hydrodynamic mode**, even for perturbations on a background that is **at rest**. But such a mode is **stable** and **decays exponentially to zero within a time scale**  $\tau_\pi$  – the new transport coefficient introduced when constructing transient theories of fluid dynamics. The other mode is hydrodynamical and, at small wavenumbers, is identical to the one obtained from relativistic Navier-Stokes theory. Therefore, at small wavenumbers, the transient theory is very similar to Navier-Stokes theory, the only difference being the appearance of a **damped** – and therefore **stable** – non-hydrodynamic mode  $\sim i/\tau_\pi$ .

At large values of the wavenumber,  $k \rightarrow \infty$ , the solution becomes

$$\omega(k \rightarrow \infty) = \pm \sqrt{\frac{\tau_\eta}{\tau_\pi}} k + \mathcal{O}(1) . \quad (2.99)$$

We see that this mode is **linear** in the wavenumber and, contrary to Navier-Stokes theory, the order of the dispersion relation in powers of frequency will **not be modified by a Lorentz boost**. As discussed, this asymptotic behavior of the mode may provide a **causal and stable mode**.

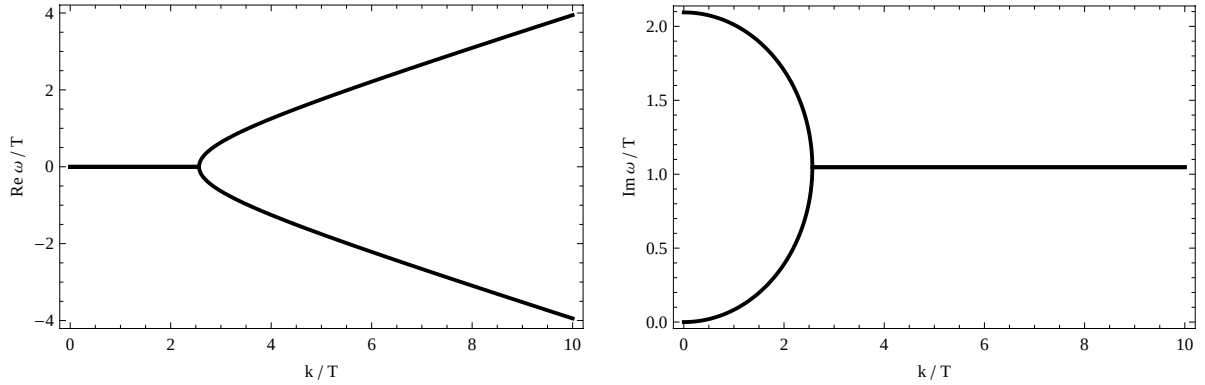


Figure 2.2: The real parts (left panel) and the imaginary parts (right panel) of the dispersion relations for the transverse modes in a static background. We fixed the relaxation time as  $\tau_\pi = 6\tau_\eta$  and  $\eta/s_0 = 1/(4\pi)$ . Figure taken from Ref. [5].

In Fig. 2.2 we show the real (left panel) and imaginary (right panel) parts of the transverse modes in units of temperature. In this plot, we parametrize the relaxation time as  $\tau_\pi = 6\tau_\eta$  [7] and fix the shear viscosity to be  $\eta/s_0 = 1/(4\pi)$ , a value usually associated to strongly coupled conformal fluids [25].

The dispersion relations for the shear modes resulting from Eq. (2.96) change their behavior from non-propagating to propagating at the critical wavenumber (2.97), as shown in Fig. 2.2. In this sense, it may be instructive to address the issue of causality by looking at the **group velocity** of such propagating modes. For wavenumbers larger than  $\kappa_c^{\text{shear}}$ , the (modulus of the) group velocity of the propagating shear mode is

$$v_g(k) = \frac{\partial \text{Re } \omega(k)}{\partial k} = v_{g,\text{shear}}^{\text{as}} \frac{k/\kappa_c^{\text{shear}}}{\sqrt{(k/\kappa_c^{\text{shear}})^2 - 1}} , \quad (2.100)$$

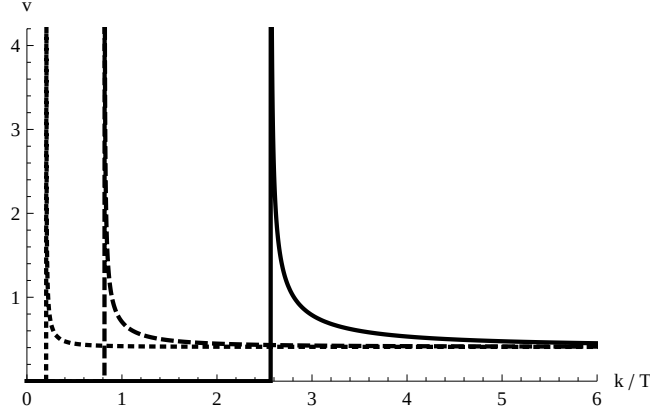


Figure 2.3: The group velocity for the transverse mode for  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = 1/3$ , and  $\eta/s_0 = 1/(4\pi)$  (full line),  $1/4$  (dashed line), and  $1$  (dotted line). Figure taken from Ref. [5].

where we defined the **asymptotic group velocity** of this mode,

$$v_{g,\text{shear}}^{\text{as}} \equiv \sqrt{\frac{\tau_\eta}{\tau_\pi}}, \quad (2.101)$$

as the asymptotic value of  $v_g$  in the large-wavenumber limit. The group velocity as a function of the wavenumber is shown in Fig. 2.3.

We can see from Eq. (2.100) (and also from Fig. 2.3) that the group velocity diverges near the critical wavenumber  $\kappa_c$  and approaches its asymptotic value ( $k \rightarrow \infty$ ) from above. For certain choices of shear relaxation time and viscosity, it may thus happen that the group velocity becomes **superluminal**. In Sec. 2.5.4 we shall show that this apparent violation of causality of the group velocity does **not** cause the theory **as a whole to become acausal**. The necessary condition required in order to insure that the theory is causal is actually that the **asymptotic group velocity** (2.101) **does not exceed the speed of light**,

$$\lim_{k \rightarrow \infty} \frac{\partial \text{Re } \omega(k)}{\partial k} = v_{g,\text{shear}}^{\text{as}} \leq 1. \quad (2.102)$$

This condition, which we shall refer to as **asymptotic causality condition**, is not automatically satisfied by the linearized Israel-Stewart theory. It actually must be **imposed additionally**, leading to the following constraint that must be satisfied by the shear relaxation time (if the effects of bulk-viscous pressure and net-charge diffusion were included, also additional constraints for the relaxation times related to these quantities would have been obtained [4, 19]),

$$\frac{\tau_\eta}{\tau_\pi} \leq 1 \quad \Longleftrightarrow \quad \tau_\pi \geq \frac{\eta}{\varepsilon_0 + P_0}. \quad (2.103)$$

That is, the relaxation time must be larger than the time scale  $\tau_\eta$ .

When analyzing the longitudinal (sound) modes, we shall find that the causality condition (2.103) is actually not sufficient to guarantee the stability and causality of the sound modes. Since all modes must be causal and stable, the validity of Eq. (2.103) does not guarantee the causality of the linearized theory as a whole.

### 2.5.2 Longitudinal modes in the rest frame

The equations of motion (2.90) – (2.92) for the longitudinal modes can be cast into the following matrix form

$$\begin{pmatrix} \Omega & -\kappa & 0 \\ -c_s^2 \kappa & \Omega & -\kappa \\ 0 & -i\tau_{\text{eff}} \kappa & 1 + i\tau_\pi \Omega \end{pmatrix} \begin{pmatrix} \frac{\delta \tilde{\varepsilon}}{\varepsilon_0 + P_0} \\ \frac{\delta \tilde{u}_\parallel}{\varepsilon_0 + P_0} \\ \frac{\delta \tilde{\pi}_\parallel}{\varepsilon_0 + P_0} \end{pmatrix} = 0, \quad (2.104)$$

with the dispersion relations being given by

$$(\Omega^2 - c_s^2 \kappa^2) (1 + i\tau_\pi \Omega) - i\tau_{\text{eff}} \Omega \kappa^2 = 0. \quad (2.105)$$

Naturally, this equation will have three solutions in both the moving background and the static one, since its highest power in frequency,  $\Omega^3$ , is the same as its highest power in both  $\kappa$  and  $\Omega$ , which is  $\Omega \kappa^2$ .

---

**Exercise 2.8:** Prove Eq. (2.105) by computing the determinant of the coefficient matrix in Eq. (2.104).

---

When the unperturbed fluid is at rest, the dispersion relation becomes

$$(\omega^2 - c_s^2 k^2) (1 + i\tau_\pi \omega) - i\tau_{\text{eff}} \omega k^2 = 0. \quad (2.106)$$

The longitudinal modes are the solutions of a cubic equation, which cannot be expressed in a simple analytical form. In the following, we initially restrict our discussion to the limiting form of these modes for  $k \rightarrow 0$  and  $k \rightarrow \infty$ , respectively. The analytic solution in the limit of small wavenumber  $k$  is

$$\omega(k) = \begin{cases} \frac{i}{\tau_\pi}, \\ \pm c_s k + i \frac{\tau_{\text{eff}}}{2} k^2. \end{cases} \quad (2.107)$$

The first solution is obtained by setting  $k \equiv 0$  in Eq. (2.106), while the second is obtained by assuming that  $\omega \sim k \ll \tau_\pi^{-1}$ , thus mapping Eq. (2.106) onto Eq. (2.78).

On the other hand, for large wavenumber one obtains from Eq. (2.106)

$$\omega(k) = \begin{cases} \frac{i}{\tau_\pi} \left[ 1 + \frac{\tau_{\text{eff}}}{\tau_\pi c_s^2} \right]^{-1}, \\ \pm c_s k \sqrt{1 + \frac{\tau_{\text{eff}}}{\tau_\pi c_s^2}} + \frac{i}{2\tau_\pi} \left[ 1 + \frac{\tau_\pi c_s^2}{\tau_{\text{eff}}} \right]^{-1}. \end{cases} \quad (2.108)$$

This corresponds to one non-propagating mode and two propagating (sound) modes. As observed for the transverse modes, at small wavenumber the longitudinal propagating



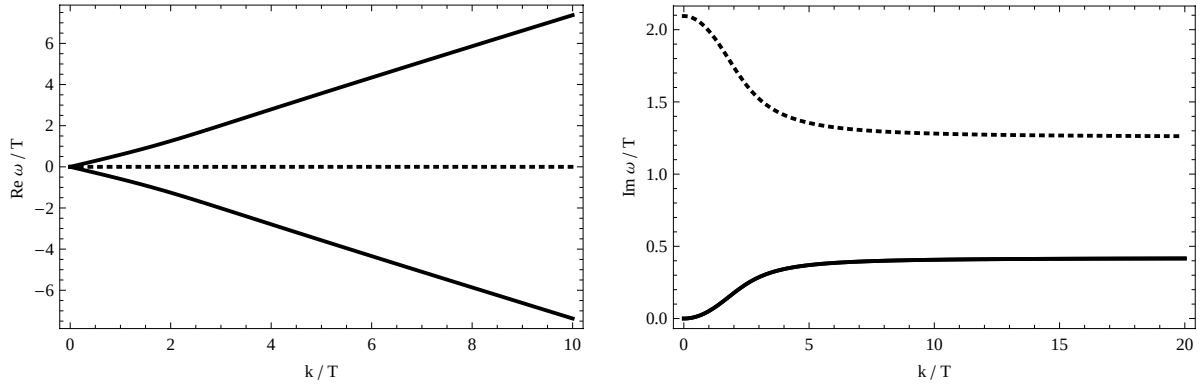


Figure 2.4: The real parts (left panel) and the imaginary parts (right panel) of the dispersion relations for the sound modes (full lines) and the non-propagating mode (dashed line). The parameters are  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = 1/3$ . Figure taken from Ref. [5].

modes are exactly the same as the ones obtained from Navier-Stokes theory. Also, we note that all imaginary parts are positive and therefore all longitudinal modes are stable (as it was the case for the transverse modes).

---

**Exercise 2.9:** Prove Eq. (2.108).

(Hint: For the first solution, assume  $\omega^2 \ll c_s^2 k^2$  in the first term of Eq. (2.106). The second solution can be obtained via perturbation theory. To this end, first consider Eq. (2.106) to leading order in  $\omega$  and  $k$ . Show that the solution of the leading-order equation gives the first term in Eq. (2.108). Then consider corrections to this solution. Linearize Eq. (2.106) in these corrections and determine them in the limit of large wavenumber. This gives the second term in Eq. (2.108).)

---

The non-propagating mode does not have a real part,  $\text{Re } \omega = 0$ , and, hence, we cannot discuss the causality of this mode using the group velocity. This is similar to what happened in Navier-Stokes theory, where all modes became purely imaginary above a critical wavenumber. For this reason, we analyze this mode by comparing it to the mode of the diffusion equation (1.113) – the same procedure adopted with Navier-Stokes theory. As already stated, the diffusion equation is known to be acausal and has a single non-propagating mode with dispersion relation  $\omega = iDk^2$ . We have already demonstrated that such a  $k^2$ -dependence in any non-propagating mode (even if in the limit of infinite wavenumber) will lead to unstable non-hydrodynamic modes once boosted to a moving frame. However, the non-propagating mode (2.108) is **independent** of  $k$  in the asymptotic limit (cf. Fig. 2.4) and should not lead to acausal signal propagation. Furthermore, this lack of dependence on the wavenumber leads to a dispersion relation that is well behaved under Lorentz transformations, i.e., no additional mode will appear when  $k^\mu$  is boosted.

The dispersion relations resulting from Eq. (2.106) are shown in Fig. 2.4, and the

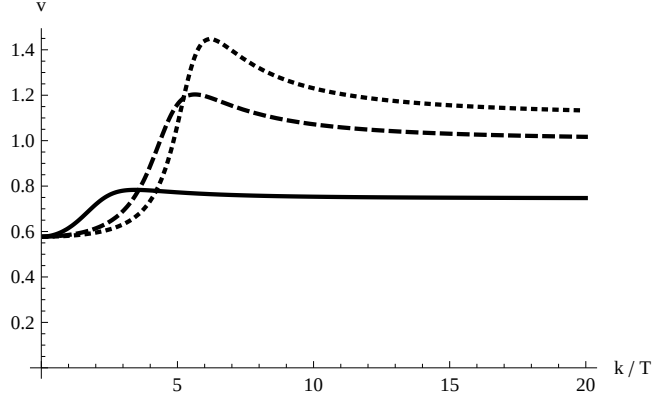


Figure 2.5: The group velocity of longitudinal modes for  $\eta/s_0 = 1/(4\pi)$ ,  $c_s^2 = 1/3$ , and  $\tau_\pi = 6\tau_\eta$  (full line),  $\tau_\pi = 2\tau_\eta$  (dashed line), as well as  $\tau_\pi = 3\tau_\eta/2$  (dotted line). Figure taken from Ref. [5].

corresponding group velocity in Fig. 2.5. The group velocity has a maximum for a finite value of  $k/T$  and approaches its asymptotic value ( $k \rightarrow \infty$ ) from above. But it does not diverge for any finite value of  $k$ , as happened with the group velocity for the shear modes. We see that for small values of the ratio  $\tau_\pi/\tau_\eta$  the group velocity can become superluminal in some wavenumber domains. Nevertheless, as will be discussed in Sec. 2.5.4, this is not necessarily a problem since only the asymptotic value of the group velocity determines whether the mode as a whole violates causality or not. However, if  $\tau_\pi/\tau_{\text{eff}}$  is sufficiently small, even the asymptotic group velocity can become larger than the velocity of light. As a matter of fact, the **asymptotic value of the group velocity for the longitudinal modes** is

$$v_{g,\text{sound}}^{\text{as}} = \lim_{k \rightarrow \infty} \frac{\partial \text{Re } \omega}{\partial k} = c_s \sqrt{1 + \frac{\tau_{\text{eff}}}{\tau_\pi c_s^2}}. \quad (2.109)$$

Consequently, for the asymptotic group velocity of sound waves to be less than the speed of light,  $\tau_\pi$  and  $\tau_{\text{eff}}$  should satisfy the following **asymptotic causality condition** [14]:

$$\frac{\tau_{\text{eff}}}{\tau_\pi} \leq 1 - c_s^2. \quad (2.110)$$

This is similar to the causality condition for the group velocity in the case of bulk viscosity, found originally in Ref. [4]. For conformal fluids, where  $c_s^2 = 1/3$ , the condition (2.110) simplifies to

$$\frac{\tau_{\text{eff}}}{\tau_\pi} \leq \frac{2}{3}. \quad (2.111)$$

Since for conformal fluids  $\zeta = 0$ , we also have  $\tau_{\text{eff}} = 4\tau_\eta/3$ , and thus

$$\frac{\tau_\eta}{\tau_\pi} \leq \frac{1}{2}, \quad (2.112)$$

which is more restrictive than the asymptotic causality condition (2.103) for the shear modes.

### 2.5.3 Stability for a moving background

So far, we have studied the dispersion relations of transient fluid dynamics for perturbations on a fluid at rest. In this scheme, it was demonstrated that such theories are **stable**, and also **causal**, if the relaxation times satisfy a certain **asymptotic causality condition**. Now we must check if this situation changes when we consider perturbations in a moving (or Lorentz-boosted) background – in Navier-Stokes theory, this completely changed the behavior with the appearance of unstable non-hydrodynamic modes.

We consider the same moving background used when analyzing the longitudinal and transverse modes of Navier-Stokes theory, which we parametrized as  $u_0^\mu = \gamma(1, V, 0, 0)^T$  and  $k^\mu = (\omega, k, 0, 0)^T$ . This corresponds to fluctuations traveling parallel to the background velocity. The dispersion relation for the transverse modes, Eq. (2.94), transforms to

$$(\omega - Vk) [1 + i\gamma\tau_\pi(\omega - Vk)] - i\tau_\eta\gamma(\omega V - k)^2 = 0. \quad (2.113)$$

where we used Eqs. (2.85) and (2.86). This is a quadratic equation and, consequently, it has two solutions, each with a twofold degeneracy. These are

$$\omega_\pm(k) = \frac{1}{2(\tau_\pi - V^2\tau_\eta)\gamma} \left[ i - 2\gamma(\tau_\eta - \tau_\pi)kV \pm \sqrt{-1 - 4i\gamma^{-1}\tau_\eta V k + 4\gamma^{-2}\tau_\eta\tau_\pi k^2} \right]. \quad (2.114)$$

---

**Exercise 2.10:** Prove that Eq. (2.114) is the solution of Eq. (2.113).

---

At  $k = 0$ , it is straightforward to see that the two solutions simplify to

$$\omega(k=0) = \begin{cases} 0, \\ \frac{i}{\gamma(\tau_\pi - V^2\tau_\eta)}, \end{cases} \quad (2.115)$$

which agree with the solution (2.98) in a non-moving background ( $V = 0$ ) and with the results obtained from Navier-Stokes theory in a moving background, Eq. (2.74), when  $\tau_\pi$  is set to zero. The main and fundamental difference is that the non-propagating mode that appears in Navier-Stokes theory, when perturbed around a moving fluid, always has a negative imaginary part. As can be seen from the expression above, it is the introduction of a shear relaxation time that makes it possible to change the sign of the imaginary part of such a non-propagating mode. As a matter of fact, we observe that, as long as  $\tau_\pi \geq \tau_\eta$ , this mode is always stable: a condition that coincides with the asymptotic causality condition (2.103). So we see once more, in a more practical setting, the connection between causality and stability.

Similarly, the dispersion relation for the longitudinal modes, Eq. (2.105), transforms to

$$[(\omega - Vk)^2 - c_s^2(\omega V - k)^2] [1 + i\gamma\tau_\pi(\omega - Vk)] - i\gamma\tau_{\text{eff}}(\omega - Vk)(\omega V - k)^2 = 0. \quad (2.116)$$

This equation is cubic and has three solutions, which are too long to write down in this context. Once more, we restrict ourselves to study the solution at  $k = 0$ , where the instability of the non-propagating modes was already obvious in Navier-Stokes theory (for a moving background). In this case,

$$\omega(k=0) = \begin{cases} \pm 0, \\ \frac{i(1 - c_s^2 V^2)}{\gamma[\tau_\pi(1 - c_s^2 V^2) - \tau_{\text{eff}} V^2]} \end{cases} \quad (2.117)$$

---

**Exercise 2.11:** Prove that Eq. (2.117) is a solution for Eq. (2.116) for  $k = 0$ .

---

This is very similar to what we found for the shear modes. We have one non-vanishing solution, corresponding to the mode that was unstable in Navier-Stokes theory, see Eq. (2.83), which agrees with the above when setting  $\tau_\pi =$ . We see once more that a **non-vanishing shear relaxation time**  $\tau_\pi$  can render this mode **stable**. For any value of  $0 \leq V < 1$ , if  $\tau_{\text{eff}} \leq \tau_\pi(1 - c_s^2)$ , the imaginary part of this mode will always be positive and, thus, it will be linearly stable. This condition corresponds to the asymptotic causality condition (2.110) derived for the longitudinal modes for a background fluid at rest, confirming that, **once causality is assured**, the boosted fluctuations will be **stable**.

In Fig. 2.6, the dependence of the group velocity on the wavenumber is shown for various values of the boost velocity  $V$ . The left panel shows the behavior of one of the shear modes and the right panel one of the sound modes. The parameter set used here is  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = 1/3$ , which satisfies the asymptotic causality condition. We observe that the divergence of the group velocity of the shear mode in the rest frame is tampered by the movement of the unperturbed fluid, resulting in a peak of finite height. However, the group velocity may still exceed the speed of light in a certain range of wavenumbers. Nevertheless, we note that as we increase the velocity of the unperturbed fluid, the peak height gradually diminishes, until the group velocity remains below the speed of light for all wavenumbers. However, the same does not occur with the longitudinal group velocity, which we found to be superluminal in some ranges of wavenumber even when the velocity of the unperturbed fluid becomes close to the speed of light.

Although the group velocity of the shear or the sound mode may exceed the speed of light, the theory is still stable as long as the asymptotic causality condition is fulfilled. This is demonstrated in the left panel of Fig. 2.7, where the imaginary parts of the modes are shown for the parameter set  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = 1/3$ . We observe that all imaginary parts are positive, indicating the stability of the theory.

In contrast to the case where the background fluid is at rest, where the theory was found to be stable even for parameters which violate the asymptotic causality condition (2.110), this is no longer the case for perturbations performed on a moving fluid. In the right panel of Fig. 2.7, the imaginary parts of the modes are calculated with the parameter set  $\eta/s_0 = 1/(4\pi)$ ,  $c_s^2 = 1/3$  and  $\tau_\pi = \tau_\eta$  – the latter violating the causality and stability conditions derived in this section. Now one observes the appearance of negative imaginary parts, indicating that the theory becomes unstable.

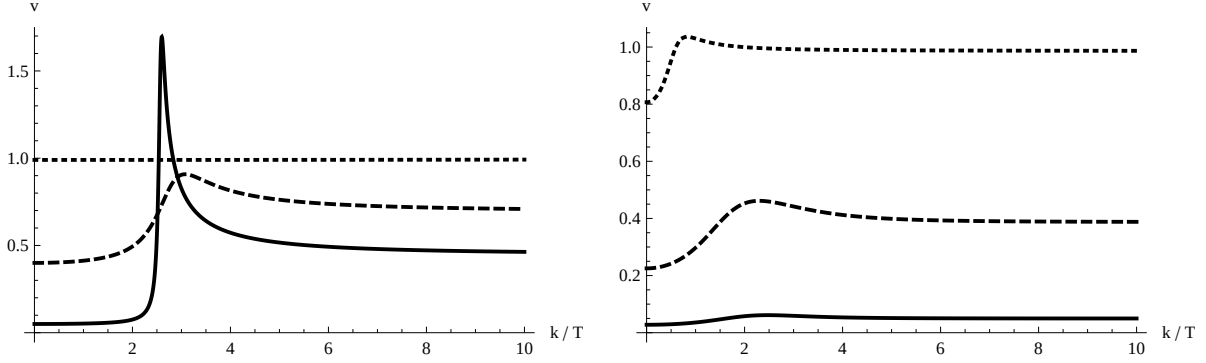


Figure 2.6: The group velocity calculated for one of the shear modes (left panel) and one of the sound modes (right panel). We set  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = 1/3$ . The solid line is for a boost velocity  $V = 0.05$ , the dashed line for  $V = 0.4$  and the dotted line for  $V = 0.99$ , respectively. Figure taken from Ref. [5].

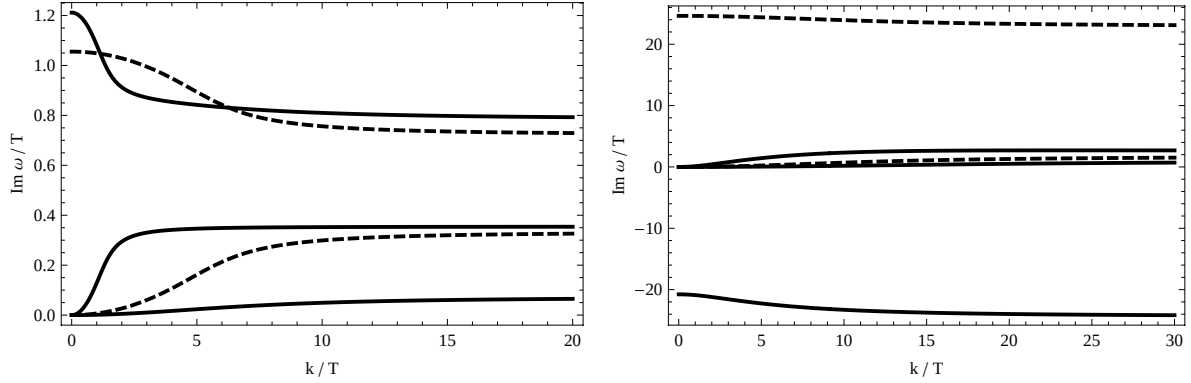


Figure 2.7: The imaginary parts of the dispersion relations for a boost in  $x$ -direction with velocity  $V = 0.9$ . The left panel shows the results for the parameter set  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = 6\tau_\eta$ ,  $c_s^2 = \frac{1}{3}$ , which fulfills the asymptotic causality condition, while the right panel is for  $\eta/s_0 = 1/(4\pi)$ ,  $\tau_\pi = \tau_\eta$ ,  $c_s^2 = \frac{1}{3}$ , which violates this condition. The dashed lines are for the shear modes, while the solid lines are for the sound modes. Figure taken from Ref. [5].

### 2.5.4 Causality of wave propagation

In the preceding discussion we have seen that transient theories of fluid dynamics can be designed to be stable as long as the transverse and longitudinal modes fulfill asymptotic causality conditions. In this section, we shall show that the **causality** of the theory **as a whole** is guaranteed if the asymptotic stability condition is fulfilled. The group velocity may become superluminal, or even diverge, as long as this apparent violation of causality is restricted to a finite range of wavenumbers. The argument leading to this conclusion is analogous to that of **Sommerfeld and Brillouin in classical electrodynamics** [26, 27]. For instance, in the case of anomalous dispersion the group velocity may become superluminal, but the causality of the theory as a whole is not affected.

The change in a fluid-dynamical variable induced by a general perturbation is given by

$$\delta X(x, t) = \sum_j \int d\omega \widetilde{\delta X}_j(\omega) e^{i\omega t - ik_j(\omega)x}, \quad (2.118)$$

where  $\delta X(x, t)$  stands for  $\delta\varepsilon$ ,  $\delta u^\mu$ , or  $\delta\pi^{\mu\nu}$ . The index  $j$  denotes the different modes, i.e., the shear modes, the sound modes etc. The function  $k_j(\omega)$  is the inverted dispersion relation  $\omega_j(k)$  of the respective mode. The Fourier components are given by

$$\sum_j \widetilde{\delta X}_j(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt \delta X(0, t) e^{-i\omega t}. \quad (2.119)$$

We assume that the incident wave has a well-defined front that reaches  $x = 0$  not before  $t = 0$ . Thus  $\delta X(0, t) = 0$  for  $t < 0$ . This condition on  $\delta X(0, t)$  ensures that  $\sum_j \widetilde{\delta X}_j(\omega)$  is **analytic in the lower half of the complex  $\omega$ -plane** [26]. On the other hand, in Secs. 2.5.1, 2.5.3 we have found that the group velocity of the shear modes **diverges** for certain values of  $k$ . These divergences correspond to **singularities in the complex  $\omega$ -plane**. However, if the asymptotic causality condition is **fulfilled**, the imaginary part of the dispersion relation is always **positive**, i.e., the singularities only appear in the **upper half of the complex  $\omega$ -plane**. In this case, the system is also **stable**. On the other hand, if the asymptotic causality condition is **violated**, the singularities may appear also in the **lower half-plane**, i.e., for **negative imaginary part** of the dispersion relation, and the system is **unstable**.

We shall now demonstrate that the divergences in the group velocity do not violate causality as long as the asymptotic causality condition is satisfied, i.e., as long as the asymptotic group velocity remains subluminal. To this end, we compute Eq. (2.118) by contour integration in the complex  $\omega$ -plane. To close the contour, we have to know the asymptotic behavior of the dispersion relations. In our calculation, we found that the real part of the dispersion relation at large  $k$  is proportional to  $k$  (see Eqs. (2.99) and (2.108)), with a coefficient which is the large- $k$  limit of the group velocity, i.e.,  $v_{gj}^{\text{as}}$ ,

$$\lim_{k \rightarrow \infty} \text{Re } \omega_j(k) = v_{gj}^{\text{as}} k. \quad (2.120)$$

Then, in the large- $k$  limit, the exponential becomes

$$\exp[i\omega t - ik_j(\omega)x] \rightarrow \exp \left[ -i \frac{\omega}{v_{gj}^{\text{as}}} (x - v_{gj}^{\text{as}} t) \right]. \quad (2.121)$$

In the case  $x > v_{gj}^{\text{as}} t$ , we have to close the integral contour in the **lower half-plane**. If the asymptotic causality condition is **fulfilled**, there are **no singularities in the lower half-plane**, and Eq. (2.118) vanishes. On the other hand, the contour should be closed in the **upper half-plane** if  $x \leq v_{gj}^{\text{as}} t$ . Then, because of the singularities, Eq. (2.118) may have a nonzero value. However, as long as we choose a parameter set for which the asymptotic group velocity  $v_{gj}^{\text{as}}$  is smaller than the speed of light, i.e., for which the asymptotic causality condition is fulfilled, **the signal propagation does not violate causality**, since the locations  $x$  where the disturbance has travelled lie within the cone given by  $v_{gj}^{\text{as}}$ , which, in turn, lies within the lightcone, q.e.d.

To conclude this section, we have shown that the **asymptotic causality condition** not only implies **stability in a general (Lorentz-boosted) frame**, but also **causality** of the theory **as a whole**.

## 2.6 Summary

In this chapter, we discussed the properties of linearized fluid-dynamical equations around global equilibrium. The main goal was to determine whether hydrodynamic fluctuations around global equilibrium are stable and causal – two fundamental properties that any physical description should satisfy. We investigated these issues for ideal fluids and the two formulations of relativistic dissipative fluid dynamics considered so far in this book: Navier-Stokes theory and transient fluid dynamics (usually represented in the form of Israel-Stewart theory).

First, we studied the linear properties of ideal fluids and recovered all its basic and well-known features. In this case, transverse modes do not appear and the longitudinal modes are purely propagating, describing the propagation of sound waves. The velocity of sound of relativistic fluids was then obtained, and given by the (square root of the) derivative of the pressure with respect to the energy density, at fixed entropy per particle (or per net-charge). All modes in ideal fluid dynamics are stable and the theory is causal, as long as the velocity of sound  $c_s \leq 1$  – a property that is satisfied by all known microscopic theories.

Next, we discussed the linear properties of relativistic Navier-Stokes theory. When investigating perturbations performed on a fluid that is initially at rest, the linear properties of relativistic Navier-Stokes theory essentially reduce to those of its non-relativistic version. In this case, transverse modes do appear and describe the diffusion of the velocity field due to viscosity. The longitudinal modes still display the propagation of sound waves, which now exhibit diffusion-type damping due to the shear and bulk viscosity of the fluid. Nevertheless, novel striking features of a relativistic version of Navier-Stokes theory appear when considering perturbations on a moving fluid. In this case, we demonstrated that unstable non-hydrodynamic modes, which simply do not exist in the non-relativistic Navier-Stokes theory, appear in both the transverse and longitudinal degrees of freedom. The existence of such unstable modes renders the general global-equilibrium solution of relativistic Navier-Stokes theory linearly unstable, posing a fundamental problem to the application of this theory to describe any relativistic fluid existing in Nature. It is this fundamental problem that the transient theories of fluid dynamics, constructed in the

previous chapter, aim to correct.

We then investigated the transverse and longitudinal modes of (second-order) transient fluid dynamics and demonstrated that such theories can be constructed to be causal and stable, at least when perturbed around global equilibrium. We argued (and later proceeded to demonstrate) that as long as the **asymptotic value of the group velocity remains subluminal**,  $v_g^{\text{as}} \leq 1$ , the theory is causal. We have shown that the condition  $v_g^{\text{as}} \leq 1$  is equivalent to the requirement that the relaxation time scale  $\tau_\pi$  must not be smaller than the ratio  $\tau_{\text{eff}}/(1 - c_s^2)$ , where  $\tau_{\text{eff}} \sim \eta/(\varepsilon_0 + P_0) \sim \beta_0 \eta/s_0$ . Thus, second-order transient theories of fluid dynamics are not *per se* stable and causal; they **may become unstable and acausal if this condition is violated**. This is an important conclusion for practitioners of fluid dynamics, who frequently consider  $\tau_\pi$  and the shear viscosity-to-entropy density ratio  $\eta/s_0$  to be independent from each other. We have demonstrated that this is not the case if one wants the theory to remain causal. These findings also illuminate, from a different perspective, why Navier-Stokes theory violates causality, because there  $\tau_\pi \rightarrow 0$ , while  $\eta$  remains nonzero. Therefore, transient theories of fluid dynamics, such as Israel-Stewart theory, can actually be successfully applied to describe the dynamics of relativistic fluids.

We finally recounted a time-honored argument from electrodynamics, proving that causality of a theory is guaranteed if the large-wavenumber limit of the group velocity remains subluminal. Thus, a divergence of the group velocity at some finite wavenumber, which actually occurs for some modes of transient fluid dynamics, does not necessarily imply that the theory as a whole becomes acausal.



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